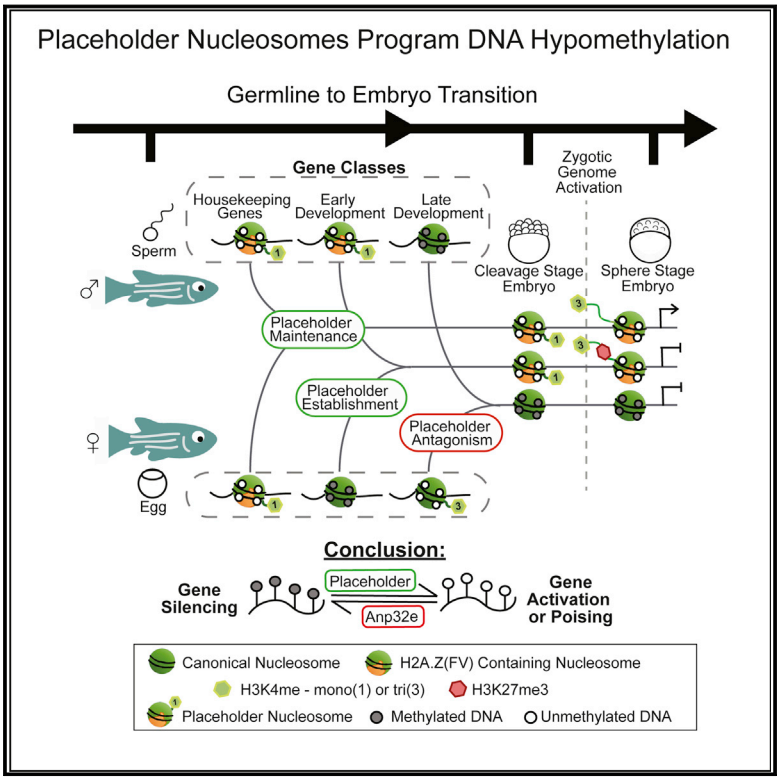
Article

Cell

Placeholder Nucleosomes Underlie Germline-to-Embryo DNA Methylation Reprogramming

Graphical Abstract



Highlights

- Distinct nucleosomes occupy all hypomethylated DNA in zebrafish sperm and early embryos
- These Placeholder nucleosomes bear H3K4me and the histone variant H2A.Z(FV)
- Upon genome activation, Placeholders become active or poised nucleosomes
- Placeholder localization excludes DNA methyla and regulates embryonic transcription

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In Brief

Transient Placeholder nucleosomes ensure that germline DNA methylation patterns set the stage for transcription during development.

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Placeholder Nucleosomes Underlie Germline-to-Embryo DNA Methylation Reprogramming

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SUMMARY

The fate and function of epigenetic marks during the germline-to-embryo transition is a key issue in developmental biology, with relevance to stem cell programming and transgenerational inheritance. In zebrafish, DNA methylation patterns are programmed in transcriptionally quiescent cleavage embryos; paternally inherited patterns are maintained, whereas maternal patterns are reprogrammed to match the paternal. Here, we provide the mechanism by demonstrating that "Placeholder" nucleosomes, containing histone H2A variant H2A.Z(FV) and H3K4me1, virtually occupy all regions lacking DNA methylation in both sperm and cleavage embryos and reside at promoters encoding housekeeping and early embryonic transcription factors. Upon genome-wide transcriptional onset, genes with Placeholder become either active (H3K4me3) or silent (H3K4me3/K27me3). Notably, perturbations causing Placeholder loss confer DNA methylation accumulation, whereas acquisition/expansion of Placeholder confers DNA hypomethylation and improper gene activation. Thus, during transcriptionally quiescent gametic and embryonic stages, an H2A.Z(FV)/H3K4me1-containing Placeholder nucleosome deters DNA methylation, poising parental genes for either gene-specific activation or facultative repression.

INTRODUCTION

A major issue in early development is how chromatin packaging of the parental genomes in embryos helps ensure proper gene regulation during and after initial genome-wide transcriptional activation. Chromatin packaging greatly differs between mature sperm and eggs, yet these differences are largely harmonized (imprinted loci excepted) via chromatin reprogramming in early vertebrate embryos (Akkers et al., 2009; Dahl et al., 2016; Hammoud et al., 2009; Liu et al., 2016; Vastenhouw et al., 2010; Wu et al., 2011; Zhang et al., 2016). The zebrafish (*Danio rerio*) model system offers an opportunity to study this reprogramming process in detail because of the abundant availability of sperm, eggs, and embryos and key properties of zebrafish cleavage-stage embryos. Unlike mammals, prior to initiating zygotic genome transcriptional activation (ZGA), zebrafish undergo 10 rapid cell divisions during cleavage phase, resulting in a transcriptionally active ~4,000-cell largely pluripotent embryo at 4 hr post-fertilization (4hpf) (Lee et al., 2013).

In zebrafish sperm and eggs, both the paternal and maternal genomes are solely packaged in histones, whereas mammalian sperm utilizes primarily protamine (Wu et al., 2011; Zhang et al., 2016). However, similar to mammals, the zebrafish sperm genome appears "poised" by chromatin for the future activation of particular gene categories. For example, zebrafish sperm lack DNA methylation at the promoters of housekeeping genes (H3K4me3-marked) and also at bivalent genes (H3K4me3 and H3K27me3-marked; Bernstein et al., 2006) that encode transcription factors (TFs) for guiding somatic development (e.g., *hox*-family, *gata*-family factors), pluripotency factors (e.g., *nanog* and *pou5f3 - zebrafish OCT4 ortholog*), and genes for germline identity/function (e.g., *dazl*, *piwil1*) (Wu et al., 2011, Potok et al., 2013). Thus, the paternally inherited DNA methylation patterns of zebrafish sperm are compatible with achieving pluripotency and with the derivation of germline. Likewise, zebrafish oocytes lack DNA methylation at housekeeping genes and certain developmental genes; however, they possess DNA methylation at several hundred loci important for development, pluripotency, and germline identity/function (e.g., many *hox*-family genes, *nanog*, *dazl*). Whereas the paternal DNA methylation patterns are virtually identical to those in pluripotent embryos at 4hpf, the maternal genome is gradually reprogrammed to the paternal pattern during cleavage (Jiang et al., 2013; Potok et al., 2013). Presently, the mechanisms that enable maintenance of paternal DNA methylation and reprogramming of maternal DNA methylation in zebrafish embryos are unknown and may inform similar issues in mammals.

In other systems, achieving DNA hypomethylation (hypoDNAme) occurs via a host of mechanisms; however, many of these cannot account for DNA methylation reprograming in early zebrafish embryos. For example, H3K4-methylation and/or H3K27me3 antagonize DNA methylation (Brinkman et al., 2012; Ooi et al., 2007; Zhang et al., 2010), yet cleavage-stage zebrafish embryos display strong focal hypoDNAme in the absence of high H3K4me3 and/or H3K27me3, which are depleted during cleavage (Lindeman et al., 2010; Vastenhouw et al., 2010; Zhang et al., 2014). Likewise, bivalent regions are

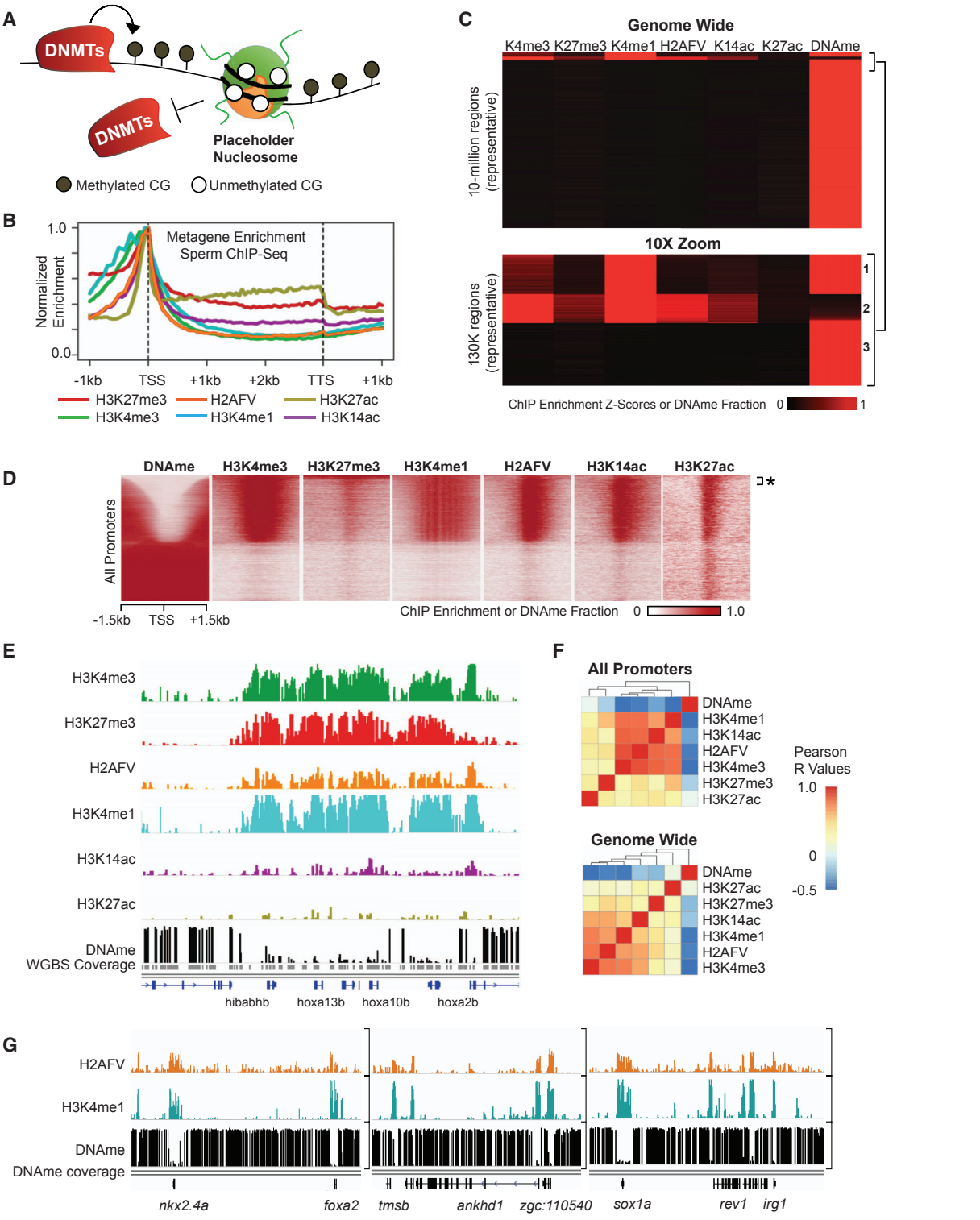


Figure 1. Identification of Placeholder Features in Zebrafish Sperm Chromatin

(A) A model for "Placeholder" nucleosome blocking DNA methyltransferases (DNMTs) to facilitate focal hypoDNAme.

(B) Metagene analysis of normalized ChIP-seq enrichment. Epigenetic features are enriched at the transcription start site (TSS) and depleted over the gene body.

(C) K-means clustering of epigenetic features in sperm. Top: representative view of the whole genome. Bottom: 10× zoom for higher resolution. *Z* scores are displayed for ChIP-seq data and fraction methylation for DNA methylation (DNAme). Regions are 200 bp.

(legend continued on next page)

also largely absent in early mouse embryos (Liu et al., 2016; Zhang et al., 2016). Thus, alternative factors or modifications may be needed in zebrafish to maintain focal hypoDNAme in early embryos. Moreover, the use of TET-family enzymes and 5hmC (Hackett et al., 2013; Huang et al., 2014; Tahiliani et al., 2009) is excluded in cleavage-stage zebrafish (Potok et al., 2013) and, instead, utilized at/after gastrulation (Bogdanović et al., 2016). Here, we provide a mechanism for establishing and maintaining focal hypoDNAme in cleavage-stage embryos, involving a nucleosome bearing the histone H2A variant H2A.Z(FV) and H3K4me1.

The histone H2A variant H2A.Z (termed H2AFV in zebrafish) is similar to canonical H2A, but differs along its amino terminal "tail" and at key residues in the globular domain and C terminus. Precise genomic incorporation of H2A.Z occurs by the nucleosome remodeler SRCAP (Snf2-related CREBBP activator protein) complex (Kobor et al., 2004; Krogan et al., 2003; Wang et al., 2016). H2A.Z installation is antagonized by its chaperone Anp32e, which promotes H2A.Z removal (Mao et al., 2014; Obri et al., 2014). In vertebrates, H2A.Z-containing nucleosomes reside at most gene promoters, including developmentally regulated genes (Hu et al., 2013; Ku et al., 2012), and H2A.Z has roles in differentiation, nucleosome stability, transcriptional regulation, and Polycomb complex recruitment (Chen et al., 2014; Creyghton et al., 2008; Hu et al., 2013; Jin et al., 2009; Ku et al., 2012). H2A.Z and DNA methylation are antagonistic, both in plants (*Arabidopsis thaliana*) and in human cancer cells (Conerly et al., 2010; Zilberman et al., 2008); however, whether H2A.Z helps maintain DNA hypomethylation during the germline-to-embryo transition is unexplored.

Here, we utilized zebrafish to address a central question in vertebrate germline-to-embryo epigenetics: how are paternal DNA methylation patterns maintained and maternal DNA methylation patterns reprogrammed during a transcriptionally quiescent cleavage phase, during which H3K4me3-marked and bivalent chromatin is absent? Our work provides evidence for "Placeholder" nucleosomes, which consist of zebrafish H2A.Z (H2AFV) and H3K4me1, to "hold in place" hypoDNAme regions, which then resolve at ZGA into either transcriptionally active or poised chromatin. Through a series of gain-of-function and loss-of-function approaches, we demonstrate the molecular utility of Placeholder in programming *in vivo* DNA methylation patterns and in the regulation of transcriptional activation during early zebrafish development.

RESULTS

Specific Chromatin Features Mark DNA Hypomethylated Regions in Sperm

DNA hypomethylated regions in sperm and in early embryos are striking in their sharp boundaries (Jiang et al., 2013; Potok et al., 2013). To account for this precision of focal DNA methylation hypomethylation during zebrafish cleavage stage, we reasoned that a "Placeholder" type of nucleosome/chromatin bearing particular histone variants and/or modifications might exist to antagonize DNA methylation in both sperm and transcriptionally quiescent embryos (Figure 1A). We identified candidate "Placeholder" marks/features by genome-wide chromatin immunoprecipitation sequencing (ChIP-seq) to profile known promoter (H3K4me3, H3K27me3, H3K14ac, and H2A.Z/FV) and enhancer features (H3K4me1 and H3K27ac) in zebrafish sperm. These marks showed the expected profiles at transcription start sites (TSS) and gene body regions (Figures 1B and S1A). K-means clustering of these data revealed low DNA methylation overlapping with high levels of these marks (Figures 1C, cluster 2, and S1B) (DNA methylation data from Potok et al. [2013]). However, DNA methylation appeared compatible with robust levels of H3K4me1 and low-moderate H3K4me3 (cluster 1), indicating their insufficiency (alone, at those levels) for deterring DNA methylation in sperm. In counter-distinction, virtually all regions with high H2AFV lacked DNA methylation and were often associated with high H3K4me1 and moderate H3K14ac (cluster 2; Figure S1C).

Similarly, the majority (55%) of sperm promoters lacked DNA methylation and were marked with one or more of these modifications, most prominently high H2AFV or H3K4me1/3 (Figure 1D). Promoters with high H3K27me3 and H3K4me3 (bivalent) also contained H2AFV and lacked DNA methylation; these represented only a small fraction (~5%) of total promoters (Figure 1D, asterisk) but included many key developmental genes (e.g., *hox* clusters; Figure 1E). Pearson correlational analyses revealed H3K4me3, H2AFV, and H3K4me1 strongly anti-correlated with DNA methylation both genome-wide and at promoters (Figure 1F), a result clearly evident in browser snapshots (Figure 1G). DNA methylation is anti-correlated with CpG density in mammals and zebrafish (Potok et al., 2013). Interestingly, nearly all sperm loci marked by H2AFV lack DNA methylation, regardless of promoter/intergenic context or CpG density (Figures S1D and S1E). Taken together, H2AFV is strikingly anti-correlated with DNA methylation and is frequently found with H3K4me1.

Acquisition of H2AFV Confers Loss of DNA Methylation in Sperm

Functional tests for H2AFV regulation of DNA methylation by *h2afv* knockout approaches were deterred due to the ubiquitous expression and separate paralogs of *h2afv*. Therefore, we disrupted the single-copy gene, *srcap*, which encodes the conserved catalytic subunit of the chromatin remodeler SRCAP complex (Kobor et al., 2004; Krogan et al., 2003; Wang et al., 2016). Our TALEN-mediated disruption (Dahlem et al., 2012) of *srcap* was successful; however, homozygous null animals failed

(D) H3K4me3, H3K4me1, H2AFV, and H3K14ac mark hypoDNAme promoters in sperm. Heatmaps of *Z* scores are displayed for ChIP-seq data and fraction methylation for DNAme. *, bivalent loci.

(E) A genome browser snapshot of the *hoxA* locus. log10 fold enrichment (FE) values (0–1) for ChIP-seq data and DNA methylation fraction (0–1).

(F) H3K4me3, H3K4me1, H2AVF, and H3K14ac are correlated with one another and anti-correlated with DNA methylation in sperm. Clustering of pairwise Pearson correlation R values are displayed as a heatmap.

(G) A gallery of browser snapshots for three genomic loci depicting the anticorrelation of H2AFV and H3K4me1 with DNAme.

See also Figure S1.

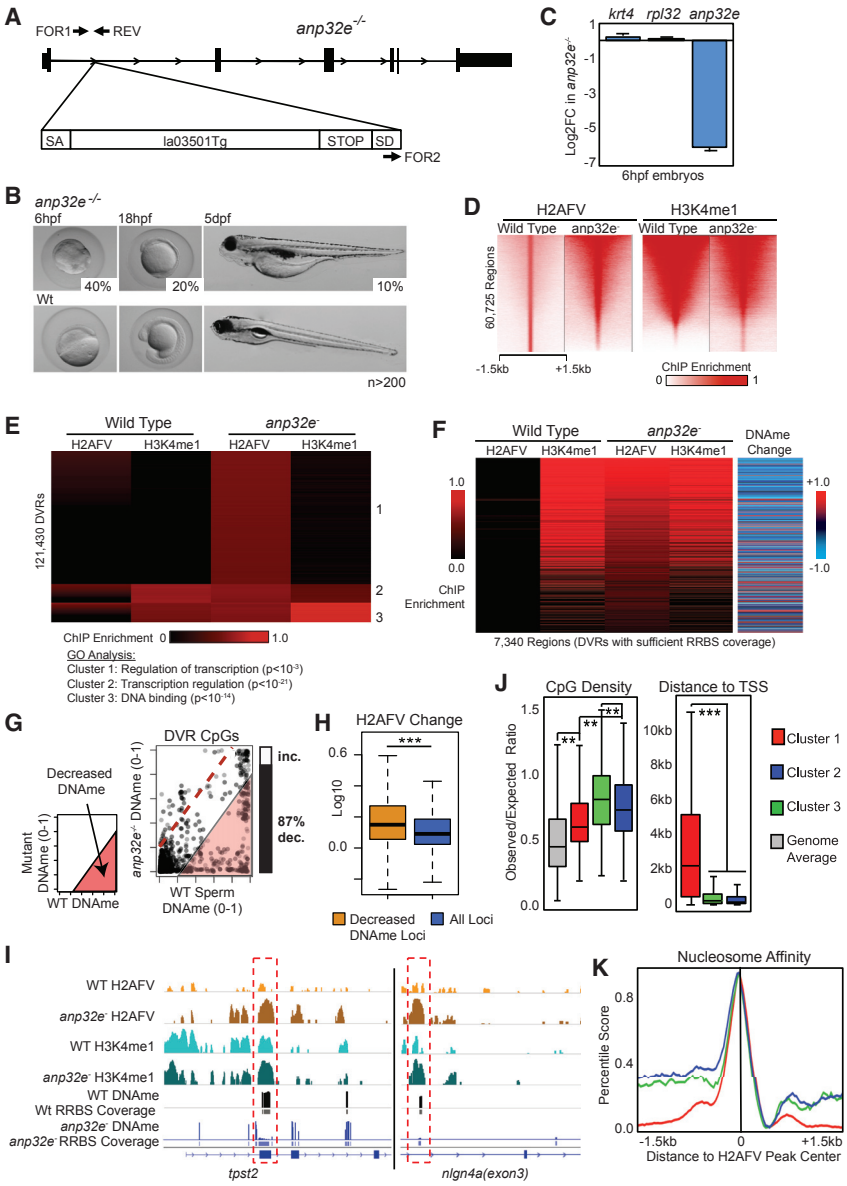


Figure 2. A Null Mutation for *anp32e* Causes Increased Genomic H2AFV and Decreased DNA Methylation in Sperm

(A) Scheme for the *anp32e* mutant allele. A gene trap cassette (la013501Tg) within the *anp32e* gene causes premature translation termination. PCR primers are indicated.

(B) *anp32e*$^{-/-}$ homozygotes display a variety of phenotypes. Lethality occurs at the displayed stages. 30% are viable and fertile.

(C) *anp32e* mRNA in the *anp32e* mutants. An early zygotically transcribed gene (*krt4*) and a housekeeping gene (*rpl32*) are unaffected. Error bars represent SD.

(D) H2AFV enrichment regions become more broad in *anp32e*$^{-}$ sperm. Heatmaps for H2AFV and H3K4me1 (normalized enrichment) are displayed at all WT sperm H2AFV peaks.

(E) *anp32e*$^{-}$ sperm display ectopic H2AFV localization. H2AFV and H3K4me1 enrichment (Z scores) is displayed as a heatmap. Differential variant regions (DVRs) were analyzed and segregate into 3 clusters by K-means clustering. Results of GO analysis for affected gene categories is shown below.

(F) Heatmaps indicate DNA methylation decreases at the majority of DVRs. Displayed regions are only those with sufficient read depth from RRBS-seq assays (see the STAR Methods).

(G) Individual CpGs within DVRs indicate decreased DNAme. Scatterplot of mutant versus control is displayed with a bar graph quantifying changing CpGs (>20% absolute change), and the percentage decreasing DNA methylation is shown.

(H) H2AFV increases more at regions at which DNA methylation decreases. Boxplots of log10FE values are shown. *** $p < 0.001$.

(I) Browser snapshots are displayed for WT and *anp32e*$^{-}$ sperm. Regions with sufficient coverage for DNA methylation data are indicated with bars beneath each track. Ectopic H2AFV accumulation is highlighted with red dashed boxes. log10FE data are from 0–1 for ChIP-seq and DNA methylation fraction from 0–1.

(J) Analysis of CpG density (left) and neighboring gene proximity distance (right) for clusters identified in (E). t tests: ** $p < 0.01$, *** $p < 0.001$. Boxplots are displayed.

(K) Analysis of predicted nucleosome affinity at DVRs. Regions from (E) were grouped by cluster and colored like in (J).

See also Figure S2.

to survive to sexual maturity (phenotypes described later). Also, transplantation of *srcap*$^{-/-}$ germline stem cells into wild-type (WT) surrogates solely yielded males that lacked sperm, consistent with a necessity for germline stem cell survival. Therefore, as an alternative strategy, we isolated zebrafish mutants lacking Anp32e, the histone chaperone known to remove H2A.Z in mice (Obri et al., 2014).

Zebrafish *anp32e* null mutants (if viable) would offer the potential to study the impact of H2A.Z mislocalization in gametes and maternal zygotic Anp32e-deficient embryos. To proceed, we obtained the available heterozygous *anp32e* null mutation (transgenic la013501 insertion; Figures 2A and S2A) (Varshney et al., 2013). Homozygous mutants displayed phenotypic variation with lethality occurring either at shield stage (~40%), early somite stage (~20%), or 5 days post-fertilization (dpf; ~10%) (Figure 2B). However, ~30% of the embryos survived to adulthood, genotyped as *anp32e*$^{-/-}$, lacked detectable *anp32e* RNA at shield stage (unlike WT) (Figure 2C; logFC = log fold change), and were fertile, providing a source of *anp32e*$^{-}$ sperm and oocytes.

We first examined H3K4me1 and H2AFV localization following Anp32e loss. Notably, we observed a clear expansion of H2AFV in *anp32e*$^{-}$ mutant sperm, which primarily occurred at intergenic regions (Figure S2B) and often involved H2AFV peak spreading into the surrounding H3K4me1-marked region (Figure 2D).

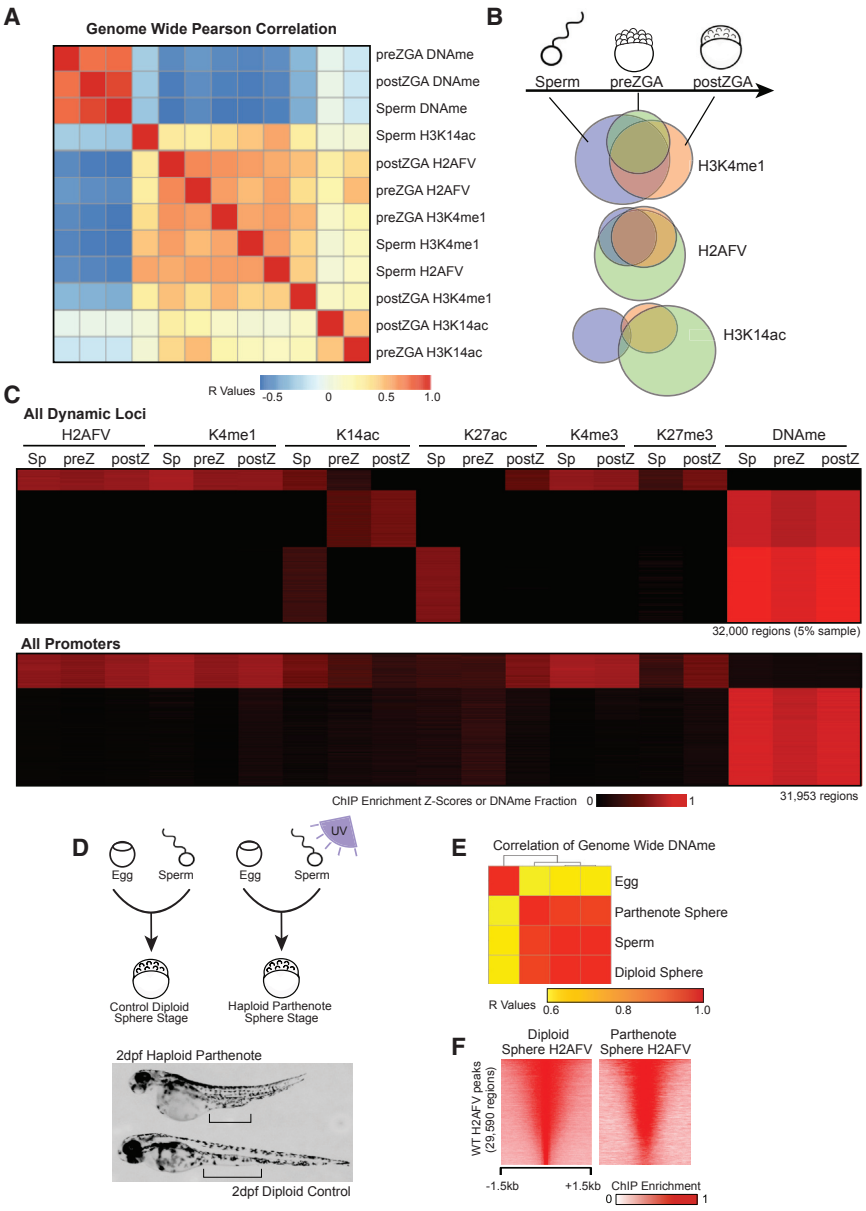


Figure 3. Maintenance of H2AFV and H3K4me1 at DNA Hypomethylated Regions in Zebrafish Embryos

(A) Correlation of Placeholder candidates in sperm and embryos. Pearson correlation R values are displayed, and the order was determined by the clustering in Figure S3C.

(B) Maintenance of Placeholder candidates during sperm-to-embryo transition. Venn diagrams indicate overlap of called peaks comparing sperm (blue), preZGA (green), and postZGA (red) samples.

(C) Heatmaps for epigenetic marks in sperm and embryos. Displayed are dynamic loci at which epigenetic marks change during development (top) and promoter regions (bottom).

(D) Haploid parthenogenesis by UV irradiation. Top: *in vitro* fertilization resulted in phenotypically WT or haploid embryos dependent on UV treatment. Bottom: 2-day-old haploid embryos were shorter with reduced intestinal length, as indicated by the black bars.

(E) DNAme reprogramming is normal in gynogenetic haploids. Genome-wide RRBS revealed DNA methylation in haploids is similar to sperm. Displayed is a heatmap for Pearson correlations.

(F) H2AFV patterns in gynogenetic embryos are normal. Haploid H2AFV at WT H2AFV peaks is highly similar to normal embryos. Displayed is a heatmap of normalized enrichment.

We further identified ~120,000 *de novo* ectopic H2AFV regions in *anp32e*$^{-}$ mutant sperm, defined as differential variant regions (DVRs; see the STAR Methods). Relationships between H2AFV and H3K4me1 at DVRs were revealed through K-means clustering, which yielded three clusters (Figure 2E): cluster 1 loci contain H2AFV but lack H3K4me1, loci within clusters 2 and 3 bear moderate-high H2AFV and H3K4me1, and cluster 3 bears the highest H3K4me1. Thus, a portion of the sites that acquire H2AFV elevate H3K4me1. Notably, examination of gene categories (via GO analysis) that acquire ectopic H2AFV yielded enrichment at genes encoding DNA-binding transcriptional regulators (Figure 2E, bottom).

Next, we quantified the impact of altering H2AFV on DNA methylation via reduced representation bisulfite sequencing (RRBS-seq). The vast majority (77%) of regions that acquired ectopic H2AFV lost DNA methylation (Figure 2F). Although most regions displayed a modest reduction (−7.4% average) in fraction DNA methylation and few gain DNA methylation (Figure 2G), the cluster which received the largest increase in H2AFV (cluster 1) showed a striking reduction in mean DNA methylation (50% to 25%). Reciprocally, regions that decreased in DNA methylation gained significant H2AFV (Figure 2H), with examples of affected loci in Figure 2I. (Note: transcription of DNMTs was not affected; Figure S2C.) DVR clusters differed in CpG content and location: cluster 1 was low CpG and intergenic, whereas clusters 2 and 3 were CpG rich and promoter proximal (Figure 2J). Thus, CpG-rich promoter-proximal regions that were marked by H3K4me1 but lacked H2AFV in WT sperm, gained H2AFV in *anp32e* mutant sperm whereas at CpG-poor distal intergenic regions that gained H2AFV in *anp32e* mutant sperm lacked H3K4me1 in WT sperm. CpG density and GC richness can affect nucleosome stability, and sequences that stabilize or destabilize nucleosomes can be predicted *in silico* (Kaplan et al., 2009). Here, we observed a high nucleosome affinity score at the center of ectopic H2AVF peaks in all three clusters (Figure 2K). Taken together, Anp32e loss causes ectopic acquisition of H2AFV in sperm at regions of high nucleosome affinity and attendant DNA methylation loss.

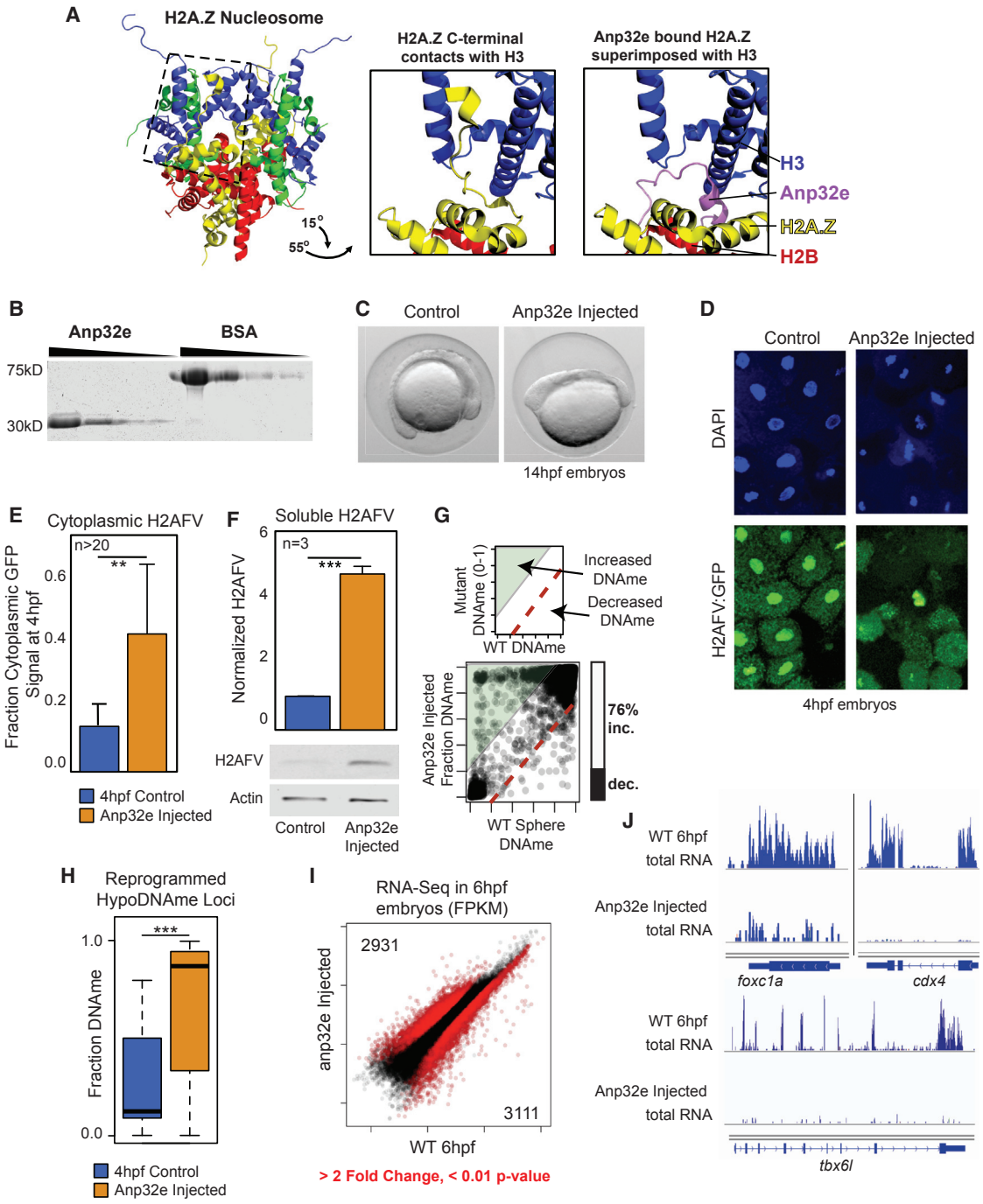


Figure 4. Anp32e Protein Injection Causes Reduced Nuclear H2AFV and Increased DNA Methylation

(A) Anp32e binding inhibits H2A.Z within the nucleosome. Left: crystal structure for the nucleosome octamer containing mouse H2A.Z (yellow) (Suto et al., 2000). Middle: zoomed-in and rotated view of the H2A.Z nucleosome octamer (dashed box from left), where the C terminus of H2A.Z interacts with histone H3. Right: the same view as middle panel, with the superimposed H2A.Z bound to human ANP32E (Obri et al., 2014).

(B) Purification of recombinant zebrafish Anp32e protein. A comparison to BSA and Coomassie staining to demonstrate purity.

(C) Phenotype of early somite-staged zebrafish following Anp32e protein injection.

(D) H2AFV signal is depleted in nuclei following Anp32e injection in embryos. DAPI and H2AFV:GFP localization in cells from sphere-staged (4hpf) embryos. Injection was into 1-cell embryos.

(E) Microscopy-based quantification of H2AFV following Anp32e injection. Fraction H2AFV:GFP signal in control and injected embryos like in (D). t tests: ** $p < 0.01$

(legend continued on next page)

Similar approaches with oocytes (either WT or *anp32e*$^{-}$) were prevented, as H2AFV levels were barely detectable, and we were unable to identify H2AFV-enriched loci in oocytes by ChIP. In keeping, H2AFV stains weakly and diffusely in oocytes (Pauls et al., 2001), suggesting minimal chromatin-bound Placeholder. Instead, we find focal hypoDNAme in oocytes co-incident with high H3K4me3 and present at housekeeping gene promoters but absent at loci for developmental TFs (e.g., *nanog*; Figure S3A and S3B). Thus, oocytes appear to utilize focal H3K4me3 to achieve focal hypoDNAme and reserve the use of Placeholder for maternal genome DNA methylation reprogramming during cleavage (this is explored below).

Locations of H2AFV in Sperm Are Largely Maintained in Embryos

Next, we compared Placeholder locations in sperm to embryos by performing genome-wide ChIP-seq in embryos for H2AFV, H3K4me1, and H3K14ac and additional modifications (Figure S3C). We performed ChIP-seq at two time points, prior to ZGA (2.5hpf) or just after ZGA (4hpf), and utilized published H3K4me1, H3K4me3, and H3K27me3 postZGA datasets (Bogdanovic et al., 2012; Zhang et al., 2014). Furthermore, we attempted ChIP-seq for H3K4me3 and H3K27me3 in preZGA samples, but, in keeping with prior observations (Vastenhouw et al., 2010; Zhang et al., 2014), these marks were very weak prior to ZGA and lacked clear ChIP-seq enrichment.

Interestingly, we find that H2AFV and H3K4me1 locations in sperm and preZGA embryos correlated and were strongly anti-correlated with embryonic DNA methylation (Figure 3A). Next, we directly compared the embryonic peak localization for H3K4me1, H2AFV, and H3K14ac to their respective peak locations in sperm. For H3K4me1, H2AFV, and H3K14ac, we observed 31%, 84%, and 7% of peak locations were maintained from sperm to preZGA embryos, respectively (Figure 3B). Here, most H3K4me1 reductions occurred at sites lacking H2AFV, and most H2AFV gains involved H2AFV spreading from existing sites into flanking H3K4me1-marked regions in embryos (Figure S3D). Most notably, the regions bound by both H2AFV and H3K4me1 in preZGA embryos lacked DNA methylation, whereas H3K14ac coincided with moderate DNA methylation (Figure 3C), reinforcing H2AFV and H3K4me1 as the consistent Placeholder marks. Supporting their co-incidence on individual nucleosomes in embryos, we co-precipitated H3K4me1 and H2AFV from mononucleosomes that were purified from preZGA embryo chromatin (2.5hpf) (Figure S3E).

Although DNA methylation patterns are nearly identical in sperm and early embryos, paternal DNA is dispensable for maternal DNA methylation reprogramming (Potok et al., 2013). Here, to determine whether embryonic H2AFV localization is also independent of paternal DNA, we generated embryos lacking sperm DNA contribution (maternal haploids) (Figure 3D, top). Our parthenogenesis procedure was highly efficient at generating phenotypically haploid 2-day-old embryos (30 of 30) (Figures 3D, bottom, and S3F) (Heier et al., 2015). As reported previously, the DNA methylation pattern of haploid parthenotes (4hpf) closely matched the sperm, and was distinct from the egg pattern (Figure 3E). Notably, H2AFV enrichment in haploid parthenotes was also highly similar to the sphere stage WT control (Figure 3F), consistent with a role for Placeholder nucleosomes in reprogramming the maternal chromatin and DNA methylation without using the paternal genome as a template, suggesting that one function of Placeholder is to establish a chromatin and DNA methylation "state" that is identical on both parental genomes.

Depletion of Nuclear H2AFV Causes Increased DNA Methylation in Embryos

To test whether Placeholder antagonizes DNA methylation in embryos, we injected translation-blocking morpholinos targeting *srcap* mRNA into fertilized 1-cell embryos or oocytes, followed by *in vitro* fertilization (IVF). *Srcap* morphants from injected 1-cell embryos died at 3 dpf, with phenotypes resembling our genetic *srcap*$^{-/-}$ mutants: an enlarged heart, a recessed jaw, and smaller retinas than WT embryos (Figures S4A and S4B). Oocyte injections improved morpholino efficacy, and conferred an earlier arrest, but only partly reduced Srcap protein in preZGA embryos (Figures S4C and S4D). This partial reduction conferred a moderate and significant increase in DNA methylation at regions normally marked by H2AFV, with higher levels at loci that normally undergo cleavage-phase loss of DNA methylation (Figure S4E). Thus, Srcap attenuation had a greater effect at regions that undergo both DNA demethylation and H2AFV acquisition in the maternal genome. Although these results support H2AFV in deterring DNA methylation, we sought an orthogonal loss-of-function approach.

The eukaryotic chaperone Anp32e antagonizes H2A.Z(FV) incorporation by binding and preventing the H2A.Z C terminus from interacting with H3 (Mao et al., 2014; Obri et al., 2014) (Figure 4A). Here, we depleted H2AFV from chromatin via ectopic elevation of Anp32e protein in embryos by injecting purified recombinant Anp32e into 1-cell embryos (Figure 4B). This approach conferred arrest at or prior to early somite stages of development in all injected animals (Figure 4C). Utilizing a transgenic H2AFV:GFP tracer line (Pauls et al., 2001), Anp32e

(F) Quantification of H2AFV in cellular fractions resulting from Anp32e injection. Western blotting for soluble H2AFV indicates that Anp32e injection causes increased cytoplasmic H2AFV. t tests: *** $p < 0.001$.

(E and F) Error bars represent SD.

(G) Anp32e injection causes increased DNAme. Scatterplot of CpGs for treatment versus control is displayed, with a bar graph quantifying changing CpGs (>20% absolute change).

(H) Changes in DNA methylation are most significant at regions at which DNA is normally demethylated. t tests: *** $p < 0.001$. Boxplots are displayed.

(I) A scatterplot of gene transcript abundance (fragments per kilobase of million mapped reads [FPKM]) for Anp32e-injected embryos compared to control samples. Red data points indicate fold change >2 and $p < 0.01$ false discovery rate (FDR). The number of genes changing is indicated in the top-left and bottom-right corners.

(J) A gallery of browser snapshots for total RNA at developmentally regulated genes.

See also Figure S4.

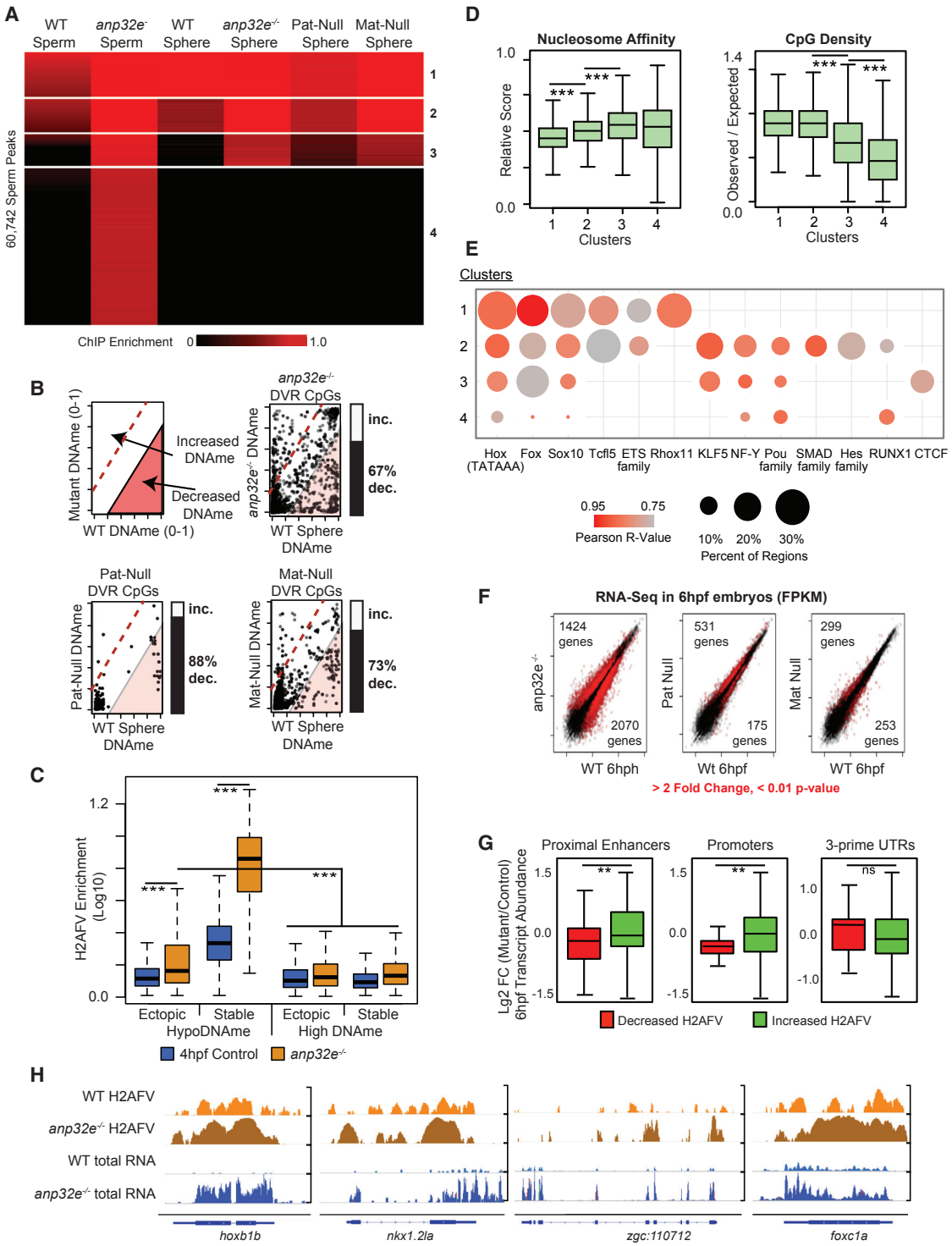


Figure 5. *anp32e* Mutation Impacts H2AFV and DNA Methylation

(A) ChIP-seq for H2AFV in sperm and sphere-stage embryos comparing WT to *anp32e* mutants. Displayed is a heatmap of K-means clustered *Z* scores at all H2AFV sperm peaks.

(B) DNAme decreases in *anp32e* mutants. Each point represents a single CpG DNA methylation for *anp32e* mutants compared to WT at differential H2AFV regions (DVRs), with a bar graph quantifying changing CpGs (>20% absolute change).

(C) H2AFV increases where DNA methylation decreases in mutants. Boxplots of H2AFV enrichment are displayed. t tests: *** $p < 0.001$.

(legend continued on next page)

injection largely relocalized cleavage phase H2AFV from the nucleus to the cytoplasm (Figure 4D), causing a 4-fold increase in cytoplasmic H2AFV (Figure 4E), which was confirmed by cell fractionation assays (Figures 4F and S4F). Thus, excess Anp32e causes considerable loss of H2AFV from nuclei. In keeping, RRBS-seq on Anp32e-injected embryos (4hpf) revealed a significant increase in DNA methylation (Figure 4G), and most dramatically at genomic regions which are normally maternally reprogrammed (Figure 4H) from high maternal DNA methylation during cleavage - similar to our observation using the *srcap* morpholino. Notably, these regions were less CpG rich than non-impacted regions and located distal to promoters (Figure S4G). Taken together, Anp32e injection caused partial nuclear depletion of H2AFV and increased DNA methylation largely at weak (CpG poor) and embryonically reprogrammed H2AFV sites.

We then determined the impact of excess Anp32e on gene transcription by RNA sequencing (RNA-seq), which yielded 3111 downregulated genes and 2931 upregulated genes (Figure 4I). Gene ontology (GO) analysis for downregulated genes identified the functional terms "DNA binding" proteins ($p < 10^{-17}$), "developmental protein" ($p < 10^{-6}$), "keratin" ($p < 10^{-5}$), and "homeobox" ($p < 10^{-4}$), whereas upregulated genes identified "transferase activity" ($p < 10^{-8}$) category (*dnmt1* not impacted; Figure S4H). Interestingly, downregulated genes were also selectively enriched for GO terms associated with development, including "dorsal/ventral pattern formation," and "Wnt signaling" ($p < 10^{-5}$), suggesting that H2AFV helps activate important developmental genes (Figure 4J).

Embryos Lacking Anp32e Have Increased H2AFV and Decreased DNA Methylation

To determine the consequence of Anp32e loss on H2AFV and DNA methylation, we compared datasets from three types of mutants: *anp32e* maternal zygotic nulls (*anp32e*$^{-/-}$), *anp32e*$^{-/+}$ maternal null zygotic heterozygotes (Mat Null), and *anp32e*$^{+/-}$ paternal null zygotic heterozygotes (Pat Null). First, in the *anp32e*$^{-/-}$ maternal zygotic null, we observed 5-fold more loci gaining H2AFV than losing H2AFV (Figure S5A), with gains primarily occurring at/around regions already bearing moderate H2AFV, and already lacking DNA methylation (Figures S5B and S5C), suggesting that Anp32e normally prevents the spreading of H2AFV in embryos, similar to our results in sperm. We then compared H2AFV sperm enrichment to H2AFV ChIP-seq data for all embryo genotypes (Figure 5A), which yielded four clusters by K-means: cluster 1 regions were occupied by high H2AFV in both WT sperm and sphere embryos. Cluster 2 regions contained moderate-to-high H2AFV in all samples, with higher levels observed when maternal Anp32e is lacking. Cluster 3 regions lacked H2AFV in WT embryos, but contained H2AFV in *anp32e* mutants, and thus appeared to rely on Anp32e to antagonize H2AFV expansion. Cluster 4 regions acquired H2AFV in sperm lacking Anp32e, but not in embryos lacking Anp32e. Thus, these loci lost H2AFV in embryos independent of Anp32e. Notably, we observed slightly less H2AFV acquisition in the Pat Null, as maternally contributed Anp32e protein likely provides some H2AFV removal. Finally, similar to our results in sperm, loss of Anp32e does not dramatically effect H3K4me1 localization (Figure S5D).

Here, DNA methylation decreased at locations at which H2AFV increased (Figure 5B), and regions of ectopic hypo-DNAme gained more H2AFV than regions with stable or increased DNA methylation (Figures 5C and S5E). In keeping, the regions where H2AFV decreased in *anp32e*$^{-/-}$ embryos (possibly from secondary/indirect effects), still displayed a significant increase in DNA methylation (Figure S5C). The clusters that gained the most H2AFV in embryos (cluster 3) or sperm (cluster 4) were distal to promoters, and exhibited lower median CpG density (but similar GC richness) and higher nucleosome affinity (Figures 5D and S5F). Taken together, Anp32e prevents H2AFV-containing nucleosomes from improperly associating with high-affinity non-promoter regions, to help ensure that these regions can acquire DNA methylation.

Next, to identify candidate factors for targeting SRCAP complex, we performed regulatory sequence analysis (Medina-Rivera et al., 2015) and identified enriched TF motifs in each cluster (Figure 5E). Cluster 1 was enriched for Homeobox and ETS family TF motifs. Cluster 2 was additionally enriched for Klf5, NF-Y, Pou family, SMAD family, Hes family, and Runx1 motifs. Clusters 3 and 4 showed diminished enrichment for these same factors, along with one unique and strong enrichment, CTCF. Interestingly, H2AFV occupancy was dramatically increased in *anp32e* mutants specifically at differentially methylated CTCF sites (Figure S5G).

Anp32e Loss Confers Activation at Genes that Gain H2AFV

Next, we determined the impact of Anp32e loss on postZGA embryo gene transcription (Figure 5F). With *anp32e* nulls, we observed 1,424 upregulated genes and 2,070 downregulated genes: upregulated genes yielded GO terms such as "homeobox" ($p < 10^{-6}$), "keratin" ($p < 10^{-5}$), and "developmental protein" ($p < 10^{-4}$), but no significant terms for downregulated genes. With heterozygotes, significant GO terms for upregulated genes were only observed when we imposed lower thresholds (greater

(D) H2AFV increases at CpG-depleted high nucleosome affinity sites in mutants. Clusters from (A) were analyzed for CpG density and predicted nucleosome affinity. t tests: *** $p < 0.001$.

(E) Transcription factor (TF) motifs differ at ectopic versus maintained H2AFV sites. Clusters from (A) were analyzed using RSAT. Motif frequency is indicated by circle size, and similarity to known binding motifs is indicated by Pearson R values in shaded red.

(F) RNA-seq on *anp32e* mutant postZGA embryos (6hpf). A Comparison of mutant to WT FPKM values. Red data points indicate FC > 2 and $p < 0.01$ FDR. The number of genes with increasing expression is indicated in the top left corner and decreasing in the bottom right corner.

(G) Increased H2AFV at promoter and enhancer regions correlate with increased transcription at *hox* genes. Strategy for parsing changes is displayed in Figure S5. t tests: ** = $p < 0.01$, ns, not significant.

(H) A gallery of browser snapshots illustrates increased H2AFV is associated with increased gene expression. ChIP-seq data are log10FE (0–1) values, and RNA-seq data are normalized read counts.

For (D and G), boxplots are displayed.

than 1.5 fold change [FC] and $p < 0.05$), all of which were developmentally associated ($p < 10^{-5}$). To investigate whether transcriptional upregulation of developmental genes was a direct result of H2AFV accumulation, we compared our H2AFV ChIP profiles to *anp32e*$^{-/-}$ expression changes for proximal genes (thresholding in Figure S5H). Notably, increased H2AFV at promoters and enhancers (but not 3′ UTRs) correlated with increased expression (Figures 5G and 5H). Taken together, loss of Anp32e increases H2AFV at the promoters and enhancers of particular developmental regulators (e.g., *hox* genes), which leads to their precocious expression following zygotic genome activation.

Notably, the GO terms for upregulated genes in *anp32e*$^{-/-}$ embryos paralleled those for repressed genes in Anp32e injected embryos, suggesting reciprocal effects on the same set of genes. Indeed, 61% of repressed gene in the Anp32e injected embryos were upregulated in *anp32e*$^{-/-}$ embryos ($p < 0.05$ and $FC > 1.5$; $p < 10^{-51}$). Remarkably, many of these genes specifically rely on DNA methylation for their proper silencing during ZGA (data from Potok et al. [2013]) (Figure 6A), including many homeobox members and genes normally expressed during gastrulation (Figure 6B), but silent at ZGA. In fact, 86% of the genes upregulated upon 5AzaC treatment (which poisons DNMTs) were repressed upon Anp32e injection ($p < 10^{-198}$), yielding a strong negative correlation (Figure 6C). As three independent methods of perturbation (Anp32e injection, *anp32e* mutation, and 5AzaC treatment) impacted transcription of a similar set of genes, these genes may normally utilize H2AFV to control their DNA methylation levels. In keeping, genes where 5AzaC caused the strongest upregulation experienced significant increases in H2AFV in the *anp32e*$^{-/-}$ embryos (Figures 6D and 6E). Taken together, intersection of these transcriptome datasets strongly suggests that Anp32e regulates/antagonizes Placeholder localization to ensure proper focal hypoDNAme and developmental gene regulation during ZGA (Figure 6F).

DISCUSSION

A central question in germline-to-embryo epigenetics in vertebrates and mammals is whether and how parental DNA methylation patterns are maintained or reprogrammed after fertilization, through the transcriptionally quiescent cleavage phase. The answers have important implications for mechanisms underlying developmental potential (totipotency or pluripotency) and transgenerational inheritance. Zebrafish and mammals both lack DNA methylation at particular gene classes in their paternal germlines, including germline-specific genes, housekeeping genes – and interestingly, also at hundreds of TFs important for embryo development (Hajkova et al., 2008; Hammoud et al., 2009; Jiang et al., 2013; Potok et al., 2013; Seisenberger et al., 2012; Vincent et al., 2013). These gene classes are packaged in two distinctive types of chromatin in zebrafish sperm: the recently ''active'' meiosis and housekeeping gene promoters are packaged in chromatin containing H3K4me1/2/3 and acetylation, whereas the ''silent'' promoters for embryo development are packaged in bivalent chromatin (H3K4me3 with H3K27me3) with H2AFV and lack acetylation. However, H3K4me3 and H3K27me3 are not retained from the gametes in the early embryo (Akkers et al., 2009; Liu et al., 2016; Vastenhouw et al., 2010; Zhang et al., 2016). Therefore, a key issue is how the DNA methylation patterns of the sperm are maintained in the embryo during a period that lacks many of the known deterrents of DNA methylation.

Regarding the maternal genome, zebrafish oocytes lack DNA methylation at housekeeping genes and certain developmental genes (those coincident with high H3K4me3), but, in contrast to sperm, oocytes bear DNA methylation at hundreds of loci important for embryo development and germline identity. Whereas the DNA methylation patterns in zebrafish sperm are virtually identical to pluripotent sphere stage embryo patterns, the maternal genome is reprogrammed gradually during cleavage such that DNA methylation patterns reflect the paternal genome (Jiang et al., 2013; Potok et al., 2013). Here, our combined genomic and functional evidence provide a mechanism for these observations, involving a Placeholder nucleosome—consisting of the histone variant H2A.Z(FV) and H3K4me1—that harmonizes the DNA methylation patterns of the two parental genomes to the same DNA methylation state and enables proper transcriptional regulation in early embryos.

We begin by discussing the conceptual issue: why utilize a Placeholder nucleosome during the germline-to-embryo transition to passively deter DNA methylation, and why not simply retain the parental marks? First, since the epigenetic marking of the two parental/gametic genomes is highly different, retaining parental marks does not allow for harmonization of the two parental epigenomes in the embryo. Second, as zebrafish do not express TET proteins (or AID) during cleavage stage, no known mechanism for targeted active demethylation appears present. Since cleavage lasts ten cell/replication cycles, a nucleosome that passively deters DNA methylation presents a simple and logical alternative. Furthermore, although H3K4me3 (or bivalency) deters DNA methylation, H3K4me3 is typically linked to transcription initiation; as cleavage stage generally lacks transcription, transcription-independent surrogate marks are logically needed to deter DNA methylation. Thus, the Placeholder nucleosome provides a mechanism both for maintaining the vast majority of the paternal regions lacking DNA methylation, and enabling reprogramming of maternal DNA methylation patterns. Regarding the name ''Placeholder,'' during a cleavage phase Placeholder occupies DNA hypomethylated genes ''in-place-of'' the other regulatory chromatin, which at ZGA will resolve into either active (housekeeping genes) or poised/silent (developmental genes) chromatin regions, both of which are DNA hypomethylated.

Our work also addresses how Placeholder nucleosomes might be targeted, as particular transcription factor (TF) motifs are highly enriched at Placeholder-occupied sites (Figure 5E). These TFs are predicted to function on both the maternal and paternal genomes: they recruit MLL complexes and SRCAP complex to maintain Placeholder locations in the paternal genome, and to install Placeholder into the maternal genome (Figure 7). This explains why the maternal genome does not need to use the paternal genome as a ''template'' to reprogram the maternal epigenome – reprogramming is a transcription factor-driven mechanism imposed on both genomes to harmonize the maternal and paternal epigenomes to the same state prior to

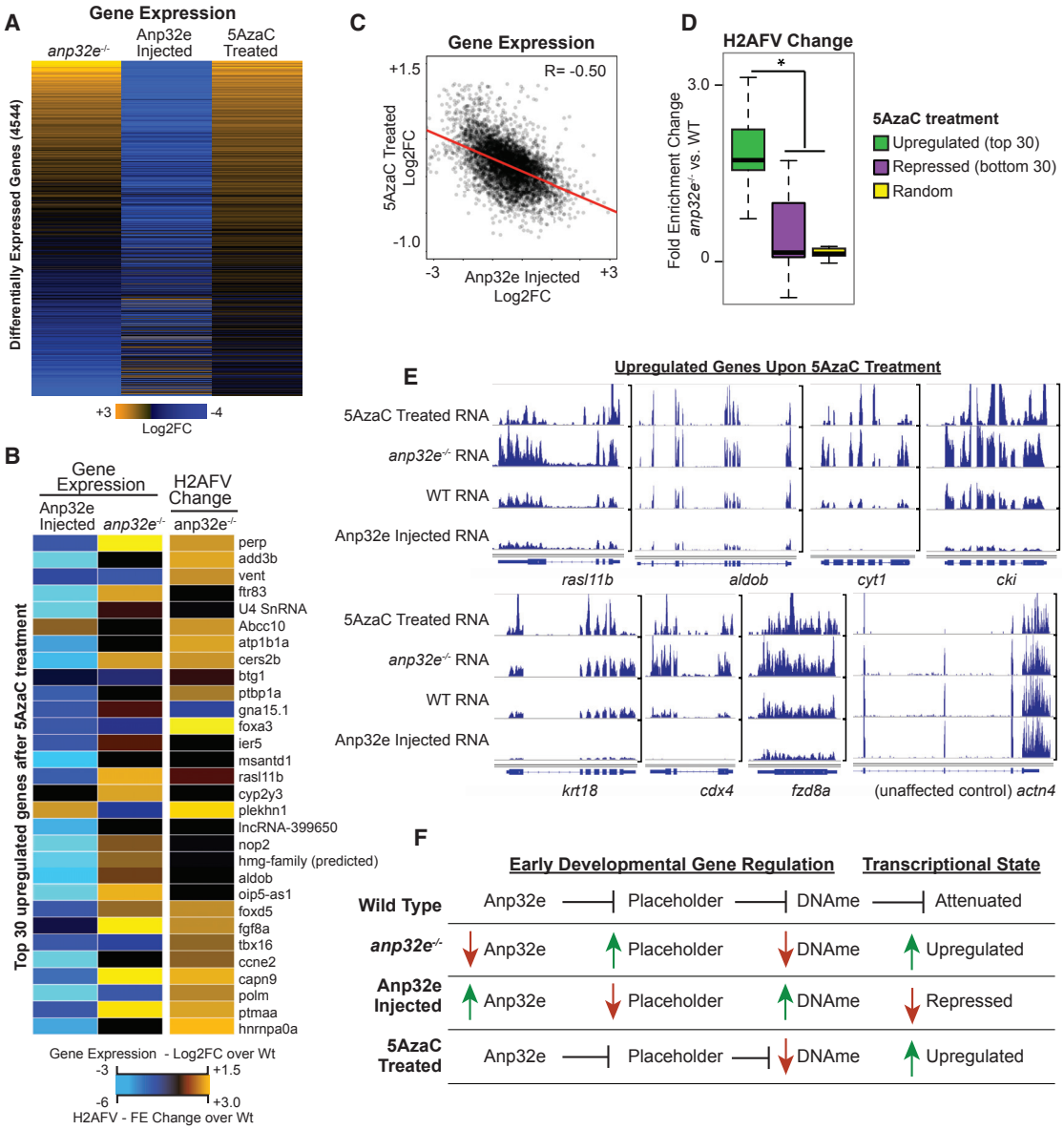


Figure 6. *anp32e* Null Mutation Causes Inappropriate Activation of Developmental Genes

(A) The same set of genes is differentially expressed in *anp32e*$^{-/-}$, Anp32e-injected, and 5AzaC-treated embryos. log2FC for differentially expressed genes (absolute FC > 1.5; minimum FPKM > 10).

(B) Genes repressed by DNA methylation are also regulated by Anp32e and marked by H2AFV. The top 30 upregulated gene from AzaC treatment are ranked by change in expression, and a heatmap indicates either gene expression change or H2AFV change.

(C) Genes downregulated in Anp32e-injected embryos are upregulated upon 5AzaC treatment. Scatterplot of log2FC is displayed for genes in (A). The Pearson R value is indicated in the upper right corner, and a red line indicates the best fit for a linear regression.

(D) H2AFV increases at promoters of genes that are regulated by DNAme. Change in H2AFV in *anp32e*$^{-/-}$ embryos is displayed for promoters (±1 kb TSS) of the top 30 5AzaC-upregulated genes and compared to repressed or to a randomly assigned list of genes. t test: * indicates p = 0.07 (n = 30). Boxplots are displayed.

(E) A gallery of browser snapshots for total RNA at genes that are repressed by DNAme, activated upon increased H2AFV. Displayed are normalized read count RNA-seq datasets.

(F) A regulatory diagram of the molecular pathway whereby Anp32e regulates Placeholder localization, which in turn regulates DNA methylation and transcriptional activation of developmental genes. Red and green arrows indicate changes at each step of the pathway based on our perturbation.

ZGA. Logically, the maternal epigenome may require extensive reprogramming due to the transcriptional processes needed for maternal RNA and protein contribution/loading in the embryo, whereas the paternal epigenome does not.

Beyond initial targeting, our work also addresses how Placeholder locations are limited and confined by the chaperone Anp32e: Placeholder nucleosomes greatly prefer nucleosome "high-affinity" sites, and Anp32e functions to "prune" H2AFV

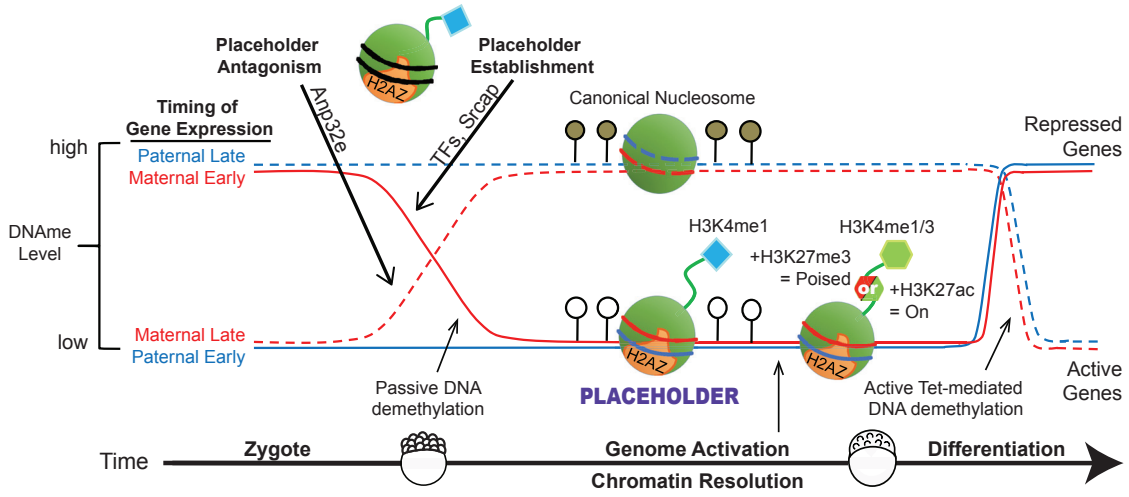


Figure 7. Model for Placeholder Regulation of DNA Methylation Reprogramming

DNAme reprogramming during cleavage phase relies on both establishment and antagonism of Placeholder nucleosomes to enable proper transcriptional activation. Placeholder is established at regions gradually undergoing maternal DNA demethylation by transcription factors and SRCAP complex, and Placeholder is antagonized at regions undergoing default acquisition of DNAme. This mechanism ensures that both housekeeping and many developmental genes (utilized early in development) acquire Placeholder and lack DNAme. Concurrent with zygotic transcriptional activation, H3K4me3 is acquired at activated housekeeping genes, and bivalent marks are established at "poised" developmental genes. Developmental genes expressed much later in development lack Placeholder and are DNA methylated.

See also Figure S6.

out of the genome, to ensure high H2AFV (H2A.Z) occupancy only at sites engaging in consistent TF-driven recruitment of the Srcap complex (Figures 7 and S6). Indeed, we observed improper H2AFV accumulation and precocious transcription of *hox* genes in *anp32e* mutants at ZGA. Thus, Anp32e refines H2AFV occupancy to prevent precocious transcription and ensure precise execution of ZGA.

Our work has several implications for chromatin-DNA methylation relationships in developing mammalian early embryos. First, H2A.Z protein is robustly detected in all stages of bovine pre-implantation embryos, and also detected in mouse embryos, though lowest at the 2-cell stage (Bošković et al., 2012). As the mouse maternal genome undergoes demethylation by passive mechanisms spanning several pre-implantation cell cycles (Amouroux et al., 2016; Santos et al., 2002; Smith et al., 2012), it will be of interest to determine whether a Placeholder-type nucleosome contributes to this demethylation process at particular locations. Following demethylation, the mammalian genome switches from DNA hypomethylated to methylated during the transition from pre- to post-implantation. A Placeholder-type nucleosome (perhaps with H2A.Z and H3K4me1) might also deter the acquisition of DNA methylation at enhancers, promoters, and other focal locations during this transition, in order to maintain these locations in a "poised/activatable" state. Furthermore, bivalent loci are largely absent in both zebrafish and mammalian cleavage-stage embryos (Liu et al., 2016; Vastenhouw et al., 2010; Zhang et al., 2016, 2014), and here we have shown that virtually all loci fated for bivalency in zebrafish emerge from a location marked by Placeholder. Thus, we speculate that pre-implantation mammalian embryos might utilize a nucleosome bearing H2A.Z and H3K4me1 to serve as a platform for re-establishing bivalency. In keeping, the loss of H2A.Z in embryonic stem cells (ESCs) diminishes the recruitment of PRC2 (H3K27me) and MLL (H3K4me) complexes (Hu et al., 2013). Furthermore, interplay may exist between H2AFV and H3K4me1, as H3K4me1 deters DNMT function *in vitro* (Guo et al., 2015), and H2A.Z promotes the recruitment of the MLL complexes, raising the possibility of a positive-feedback loop between H2AFV and H3K4me1 that collaborates to restrict DNMT function in embryos. Finally, although Placeholder in zebrafish is defined as H2A.Z(FV) with H3K4me1, the concept of a Placeholder-type nucleosome may be functionally conserved in other species, but utilize an alternative variant/mark combination to deter DNA methylation.

Taken together, our work provides evidence that parental DNA methylation status can be inherited, maintained or reprogrammed in the embryo through the installation of a Placeholder nucleosome during stages of transcriptional quiescence. Placeholder functions as a foundational chromatin "platform" in the early embryo, residing at all gene promoters that transition either to activity or bivalency at ZGA, and understanding how these transitions are accomplished represents an interesting area of future study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures and can be found with this article online at https://doi.org/10.1016/j.cell.2018.01.022.

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AUTHOR CONTRIBUTIONS

Experimental design: B.R.C., P.J.M., and S.F.W; experiment execution: P.J.M., S.F.W., C.R.J, and C.L.W.; data analysis and figures: P.J.M; manuscript preparation: P.J.M and B.R.C with comments from all authors.

DECLARATION OF INTERESTS

All authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
H2A.Z	Active Motif	Cat# 39113
H3K4me3	Active Motif	Cat# 39159
H3K4me1	Active Motif	Cat# 39297
H3K27ac	Abcam	Cat# ab4729
H3K27me3	Active Motif	Cat# 39155
H3K14ac	Active Motif	Cat# 39599
Srcap	Abcam	Cat# ab99408
Histone H3	Abcam	Cat# ab46765
H2A.Z	Abcam	Cat# ab4174
H3K4me1	Abcam	Cat# ab8895
Actin	Shaw Lab (University of Utah)	N/A
Experimental Models: Organisms/Strains		
Wild Type Tuebingen Zebrafish	University of Utah Zebrafish Core Facility	N/A
h2afv:GFP Zebrafish	gifted from Dr. Marnie Halpern; Pauls et al., 2001	N/A
anp32e mutant Zebrafish	Zebrafish International Resource Center	ZIRC - ZL5353.02; la013501Tg
srcap mutant Zebrafish	This study	N/A
Critical Commercial Assays		
Methylated Barcodes (RRBS)	BioO Scientific	cat# 511912
Chemicals, Peptides, and Recombinant Proteins		
Anp32e - recombinant zebrafish (MEMKKRISLELRNRTPAEVAEL VVDNCRSSDGEIEGLTDDFKELEFLSMVNVGLTSLAKLPSLPKLR KLELSDNNISGTLETLAEKCANLTYLNLSGNKIKELSTLEALQNLK NLKSLDLFNCEITTLEDYRESIFELLPQVTYLDGFDAEDNEAPDSE ADDDDDDEDGDEGAGQLGEYEEEEEEDEEGSEGGEVGLSYLMK EDIQDEEDDDDYVEEEEEEGGEEEADVRGEKRKREAEDEGEDDD EDDDHHHHHHHHHH)	This study	N/A
Srcap Morpholino (GATCCTTCCGCCATAGCGGTTTGCC)	Gene Tools	N/A
Oligonucleotides		
anp32e genotyping FOR1 (CGCAGCAATAACCAACCAAATGGA)	Integrated DNA Technologies	N/A
anp32e genotyping FOR2 (CGTCAGCGGGGGTCTTTCAT)	Integrated DNA Technologies	N/A
anp32e genotyping REV (TCCTGCTGGAAGCGGGAACAATATC)	Integrated DNA Technologies	N/A
srcap HRMA FOR1 (CTGGACCAAACCCAATGCTTT)	Integrated DNA Technologies	N/A
srcap HRMA REV1 (GGTACCTCCAAGACTTGCGT)	Integrated DNA Technologies	N/A
rpf1 RTPCR FOR1 (GCGAACACTACTGACTTTGAAATG)	Integrated DNA Technologies	N/A
rpf1 RTPCR REV1 (CCCAAGACAATAGAAAACCAGAGA)	Integrated DNA Technologies	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
dnmt1 RTPCR FOR2 (TTGAACTTTGAGTATGTGCGAGAG)	Integrated DNA Technologies	N/A
dnmt1 RTPCR REV2 (TTGGCTTCCATTGAGTCTCTTTAT)	Integrated DNA Technologies	N/A
Deposited Data		
RRBS-Seq	This study	GEO: GSE95033
RNA-Seq	This study	GEO: GSE95033
ChIP-Seq	This study	GEO: GSE95033
postZGA ChIP-Seq (H3K4me3, H3K27me3)	Zhang et al., 2014	GEO: GSE44269
postZGA ChIP-Seq (H3K4me1, H3K27ac)	Bogdanovic et al., 2012	GEO: GSE68087
Whole Genome BS-Seq	Potok et al., 2013	DRA/SRA: SRP020008
Sperm RNA-Seq	Jiang et al., 2013	GEO: GSE44075
Egg ChIP-Seq H3K4me3	Zhang et al., 2016	GEO: GSE71434
5AzaC treated postZGA RNA-Seq	Potok et al., 2013	DRA/SRA: SRP020008
Software and Algorithms		
Useq Package (v 8.8.8)	Nix et al., 2010	http://useq.sourceforge.net
GO (David 6.7)	Huang et al., 2009	https://david-d.ncifcrf.gov
Homer (v 4.8.3)	Heinz et al., 2010	http://homer.ucsd.edu/homer/
Novoalign (v 2.8)	Novocraft	http://www.novocraft.com
Cluster 3.0	Michael Eisen of Berkeley Lab	http://bonsai.hgc.jp/~mdehoon/software/cluster/software.htm
R	R Core Team, 2013	https://www.r-project.org/
Samtools (v 1.4)	Li et al., 2009	http://samtools.sourceforge.net/
Macs2 (v2.1.120160309)	Zhang et al., 2008	https://github.com/taoliu/MACS
DESeq2	Love et al., 2014	http://bioconductor.org/packages/release/bioc/html/DESeq2.html
Bio-ToolBox (v1.40)	Tim Parnell of the Huntsman Cancer Institute	https://github.com/tjparnell/biotoolbox
Java TreeView(3.0)	Saldanha, 2004	http://jtreeview.sourceforge.net/
CEAS	Ji et al., 2006	http://liulab.dfci.harvard.edu/CEAS/download.html
RSAT - peak-motifs	Medina-Rivera et al., 2015	http://rsat01.biologie.ens.fr/rsat/peak-motifs_form.cgi
DANPOS2	Chen et al., 2013	https://sites.google.com/site/danposdoc/
Picard(1.8)	Broad Institute	http://broadinstitute.github.io/picard/
UCSC Exe Utilities	UCSC Genome Browser	http://hgdownload.soe.ucsc.edu/downloads.html#source_downloads
pheatmap (v 1.0.8)	Raivo Kolde R Package	https://cran.r-project.org/web/packages/pheatmap/pheatmap.pdf

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for reagents and resources should be directed to and will be fulfilled by the Lead Contact Bradley R. Cairns (brad.cairns@hci.utah.edu).

EXPERIMENTAL MODELS AND SUBJECT DETAILS

Zebrafish Danio rerio strains were maintained under accordance with approved institutional protocols at the University of Utah (Westerfield, 2007). All experiments using zebrafish were approved by IACUC Protocol 17-01006. Wild-type zebrafish were from the Tubingen (Tu) strain. Transgenic H2AFV:GFP zebrafish were generated previously (Pauls et al., 2001) and gifted to us by Marnie Halpern. *Srcap* mutant fish were generated with assistance from the University of Utah mutagenesis core using Talen based methods

(Dahlem et al., 2012). *Anp32e* mutant fish were generated from a gene trap mutagenesis study (Varshney et al., 2013) and were acquired from the Zebrafish International Resource Center (ZIRC). Experimental samples were either mature gametes (sperm or oocytes) or early zebrafish embryos ranged from 0hpf to 6hpf. All other mutant zebrafish stages were phenotyped just prior to the stage of their developmental lethality, and we observed no sex bias in this study. Recombinant Anp32e protein was generation using *E. coli* BL21-CodonPlus(DE3)RIL bacteria.

METHODS DETAILS

ChIP-Seq for determining genomic localization of chromatin features

Sperm ChIP-Seq methods were similar to those described previously (Hammoud et al., 2009; Wu et al., 2011). Briefly, ~50 million sperm cells were treated with 0.05% lysophosphatidylcholine (Sigma-Aldrich L1381) in PBS on ice for 10 minutes. 5ul of Micrococcal Nuclease (New England Biolabs #M0247S) was then applied to release nucleosomes into the supernatant for 15 minutes at 37°C. Five micrograms of specific antibody conjugated to Dynabeads was incubated with the oligonucleosome solution overnight at 4°C to immuno-precipitate chromatin. Standard wash and elution methods were applied with washes in PBS supplemented with 0.5% Triton X-100, and elution was in RIPA (see below). For ChIP in early embryos, whole embryos were cross-linked with 2.2% formaldehyde for 15 minutes before quenching with 250mM glycine. Nuclei were purified as described previously (Potok et al., 2013) and re-suspended in IP dilution buffer (16.7mM Tris-HCL, 167mM NaCl, 1.2mM EDTA, 1.1% Triton-X, 0.1%SDS). Lysates were sonicated using a Branson Digital sonicator at 30% amplitude for 8 cycles of 10-pulse sonication (1 pulse per second) with 30 s rest on ice between each round of 10-pulse sonication. 5ug of antibody was then added to the lysate and left overnight to bind chromatin at 4°C rotating. The following morning, Protein A/G Dynabeads (ThermoFisher ID: 10003D) were bound to the lysate at 4°C for 4 hours. Samples were then washed 4 times in RIPA buffer (50mM HEPES, 500mM LiCl, 1mM EDTA, 1% NP-40, 0.7% SDS) and eluted overnight at 65C in RIPA. DNA was purified from eluate by phenol-extraction and alcohol precipitation. Libraries were made using the NEBNext ChIP-Seq Master Mix Set (New England Biolabs ID: E6240S). High-throughput sequencing was performed by Illumina's protocol for 50bp single-end runs on an Illumina HiSeq 2500.

Reduced Representation Bisulfite Sequencing (RRBS-Seq) for determining genomic localization of DNA methylation

Approximately 1000ng of DNA was purified from zebrafish embryos and digested with MspI, then end repaired using Klenow -exo (New England Biolabs ID: R0106S and M0212S). Methylated barcodes (BioO Scientific ID: 511912) were ligated to each sample using T4 DNA ligase (New England Biolabs ID: M0202S), and bisulfite treated according to manufacturer's protocol (Zymo Research ID: D5005). Following bisulfite treatment libraries were PCR amplified for 20 cycles using Platimun Taq (ThermoFisher ID: 10966018) and the standard primers and protocol from BioO Scientific methylated barcodes kit. High-throughput sequencing was by Illumina's protocol for either 50bp single end or 125bp paired end runs on an Illumina HiSeq 2500.

RNA-Seq to measure total RNA and quantify gene expression

RNA was purified by trizol-chloroform extraction (ThermoFisher ID: 15596026) of whole embryos followed by RNA-column cleanup. Briefly, 10 μL of squeezed sperm or 200 embryos were lysed in 500ul of Trizol, and following chloroform extraction, the RNA-containing aqueous phase was mixed 1:1 with 100% ethanol. Samples were bound to an RNAeasy column (QIAGEN ID: 74104) by centrifugation according to manufacturer's recommendations, which indicate efficient removal of most DNA without DNase treatment. Column bound RNA was then washed with RW1 and RPE buffers and then eluted with water. Purified RNA was then prepared as libraries for sequencing using the TruSeq Stranded Total RNA Library Prep Kit with Ribo-Zero (Illumina ID: RS-122-2201). High-throughput sequencing was by Illumina's protocol for 50bp single-end runs on an Illumina HiSeq 2500.

Isolation of pellet and supernatant fractions

The isolation method was adapted from Wysocka J et al., 2001, in order to accommodate zebrafish embryos. Briefly, 100 embryos were dechorionated with pronase (1mg/ml, Sigma, cat# 11459643001), and resuspended in deyolking buffer (55mM NaCL, 1.8mM KCl, 1.25mM Sodium Biocarbonate), followed by centrifugation at 500 x g, 4°C for 3 minutes to collect cellular portion of the embryo. Extraction of detergent soluble proteins from cell pellet was performed by treatment with 200 μL Buffer A (10mM HEPES, pH 7.9, 10mM KCl, 1.5mM MgCl2 0.34M Sucrose, 10% Glycerol, 1mM DTT, 0.1% Triton X-100, and protease inhibitor). Cells were incubated on ice for 8 minutes, then 100 μL was removed for total sample. The soluble and pellet fractions were collected by centrifugation at 1300 x g, 4°C for 5 minutes. 4xSB (SDS containing sample loading buffer) was added to the soluble fractions and boiled at 100°C for 5mins. The pellet fraction was resupsended in equal volume of 100 μL of Buffer B (3mM EDTA, 0.2mM EGTA, 1mM DTT, protease inhibitor). To solubilize the pellet fraction for further analysis it was incubated with 1000U of Micrococcal Nuclease (NEB, Cat# M0247S) at 37°C for 30 minutes and finally 4xSB (SDS containing sample loading buffer) was added and the samples were boiled at 100°C for 5 minutes. Samples were further analyzed by western blot analysis.

Live cell/embryo imaging

Approximately 5 H2AFV:GFP embryos (injected with either Anp32e protein or BSA) per batch were dechorionated with pronase (1mg/ml, Sigma, cat# 11459643001), for 5mins. The dechorionated embryos were rinsed with a 1:1 mix of 1xPBS and 1mg/ml

DAPI and placed on a glass bottom microscope slide dish with approximately 10uL of 1:1 mix 1xPBS and 1mg/ml DAPI. Cells were imaged on Nikon A1 Ti-E inverted microscope equipped with Four Photo Multipliers Tube (PMT) detector unit. Images were taken utilizing 405nM diode laser (DAPI) and 488nm Argon Gas laser (GFP) utilizing a 60x oil immersion objective. Z sections were acquired for a plane of cells within an embryo at 0.5 μm steps. Intensity measurements were calculated using ImageJ.

Mutant generation and genotyping

Germline transplantation assays were performed as described previously (Ciruna et al., 2002). Genotyping of *anp32e* mutant fish acquired from Zebrafish International Resource Center occurred by PCR amplification using two separate PCR reactions. Primers for the wild-type allele were CGCAGCAATAACCAACCAAATGGA and TCCTGCTGGAAGCGGGAACAATATC, and for the mutant allele primers were CGTCAGCGGGGGTCTTTCAT and TCCTGCTGGAAGCGGGAACAATATC. Mutation in the *srcap* gene were achieved by TALEN and genotyping of the mutant allele (which occurred within exon 13) was by PCR HRMA assays as described previously (Dahlem et al., 2012) using the following primer set: CTGGACCAAACCCAATGCTTT, GGTACCTCC AAGACTTGCGT.

Anp32e protein purification

The mRNA sequence for zebrafish Anp32e with a C-terminal 10X histidine tag was cloned into the pET28a plasmid, which was then electroporated into BL21-CodonPlus bacteria. Bacteria were then grown at 37°C in 3L of LB media until OD600 reached 0.8. Then, cultures were induced with 0.2mM IPTG and left at 16C shaking overnight to allow for protein expression. The following morning bacteria was gently pelleted, solubilized in lysis buffer (50mM Tris-HCL, 500mM NaCl, 10% glycerol, 0.1% triton-X, 1.5mM BME), and then sonicated using a Misonix sonicator at 50% power for 5 cycles of 1 minute on/off. Samples were then pelleted at 20,000 g for 10 minutes, and the soluble portion was recovered and bound to Ni-NTA resin at 4°C for 2 hours. Ni-bound protein was then washed 6X using 1mL lysis buffer supplemented with gradually increasing immidazol concentrations (5-60mM), then eluted with 4-1mL fractions of lysis buffer containing 200mM immidazol. Protein aliquots of 0.5mg/mL were stored at −80C in 25% glycerol, and diluted 1:5 into water immediately before use.

Oocyte or embryo injection and inhibitor treatment

Embryo injections for *srcap* morpholinos (GATCCTTCCGCCATAGCGGTTTGCC) were as described previously (Rai K. et al., 2006). Briefly, 4ng of denatured morpholinos was injected into zebrafish embryos at the 1-cell stage according to standard methods (Westerfield, 2007). Anp32e protein injection occurred similarly, except 0.1ng of protein was injected into the yolk of each 1-cell embryo. Oocyte injection and *in vitro* fertilization were performed using *srcap* morpholino concentrations identical to embryo injections, and oocyte handling was as described previously by others (Nair S. et al., 2013). Here, non-activated oocytes were collected and handled prior to fertilization in Sorting Medium (90% Leibovitz's pH 9.0, adjusted with 10N NaOH, 0.5% BSA and 100 μg/ml gentamycin). Following injection but prior to fertilization, injected oocytes recovered for 30 minutes at room temperature. *In vitro* fertilization of oocytes occurred in minimal volume of embryo water after removal of sorting media.

QUANTIFICATIONS AND STATISTICAL ANALYSES

Bioinformatics Analysis

Fasta files were aligned to Zv10 using the most updated version of Novocraft Novoalign with standard parameters for ChIP-Seq and RNA-Seq, and using bisulfite mode for the RRBS-Seq experiments. For ChIP-Seq, alignments were performed using Novoalign with the following parameters: -r Random -k -Q 13 -o SAM. Data was then processed using MACS2 for enrichment scoring and peak calling. Briefly, read count normalization was performed on alignment files to account for sequencing depth differences, and then background corrected based on input using the bdgdiff function of Macs2. Finally, Log10 fold enrichment over input lambda was then calculated for each nucleotide across the entire genome using the bdgcmp function of Macs2. These scores were then used for Macs2 based peak calling and for downstream bioinformatics analysis, which was performed in R using standard methods. Only the 'pheatmap' package was used to generate images of clustered Pearson correlation heatmaps. All other clustering was performed using CLUSTER 3.0, and these heatmaps were generated using Java TREEVIEW. ChIP-Seq regions were classified as peaks using Macs2 based on greater than 3-fold enrichment over input lambda. In sperm, Differential Variant Regions (DVRs) were defined as regions where mutant H2AFV increased more than 3-fold compared to wild-type, and in embryos, DVRs increased from less than 1.5-fold enrichment to greater than 4-fold enrichment (in mutants compared to wild-type). Metagene analysis and regional enrichment profiling were performed using DANPOS. Position relative peak enrichment was performed using CEAS. Identification of transcription factor motifs was performed with RSAT in ChIP-Seq mode. For RNA-Seq, alignment files were generated using Novoalign with the following parameters: -o SAM –r ALL 50. Aligned splice junction were converted to genomic coordinates and low quality and non-unique reads were further parsed using SamTranscriptomeParser (USeq; v8.8.8) under default settings. Stranded differential expression analysis was calculated with DefinedRegionDifferentialSeq which utilizes DESeq2 and the reference Zv10 Ensembl gene table downloaded from UCSC. Aligned RRBS-Seq data was processed using the USeq pipeline for bisulfite converted DNA as described previously (Potok et al., 2013) with Novoalign parameters of -o SAM –h 120 –t 240 –b 2 –R 3, and only regions with greater than 8 reads were considered. Parsing of bisulfite alignments was performed using NovoalignBisulfiteParser application and the point data was then parsed into mCG context using the ParsePointDataContexts application. BisStat was used to calculate

per base fraction methylation scores for bases with five or more reads from both strands and to convert data to useq format. Useq files can be converted into bigwig files for visualization in IGV using the USeq2UCSCBig application. For further analysis, bigwig files can be converted into bedgraph format using UCSC application.

Statistical Methods

Asterisks indicate the p values where a Welsh Two-Sample t test was performed under the assumption of unequal variance and non-paired samples. Default statistical thresholds ($p < 0.01$) were applied as part of DESeq2 to identify differentially expressed genes in RNA-Seq analysis, unless otherwise stated. For correlational analysis, a Pearson test was applied on complete datasets - regions where data was missing were excluded and R-values are indicated where applicable. Gene Ontology analysis utilized DAVID under default settings and p values are indicated within the text. For comparing RNA-Seq datasets, hypergemetric tests were applied to identify overlapping gene sets and p values are indicated.

DATA AND SOFTWARE AVAILABILITY

The accession number for the raw and partially processed data reported in this paper is GEO: GSE95033.s.

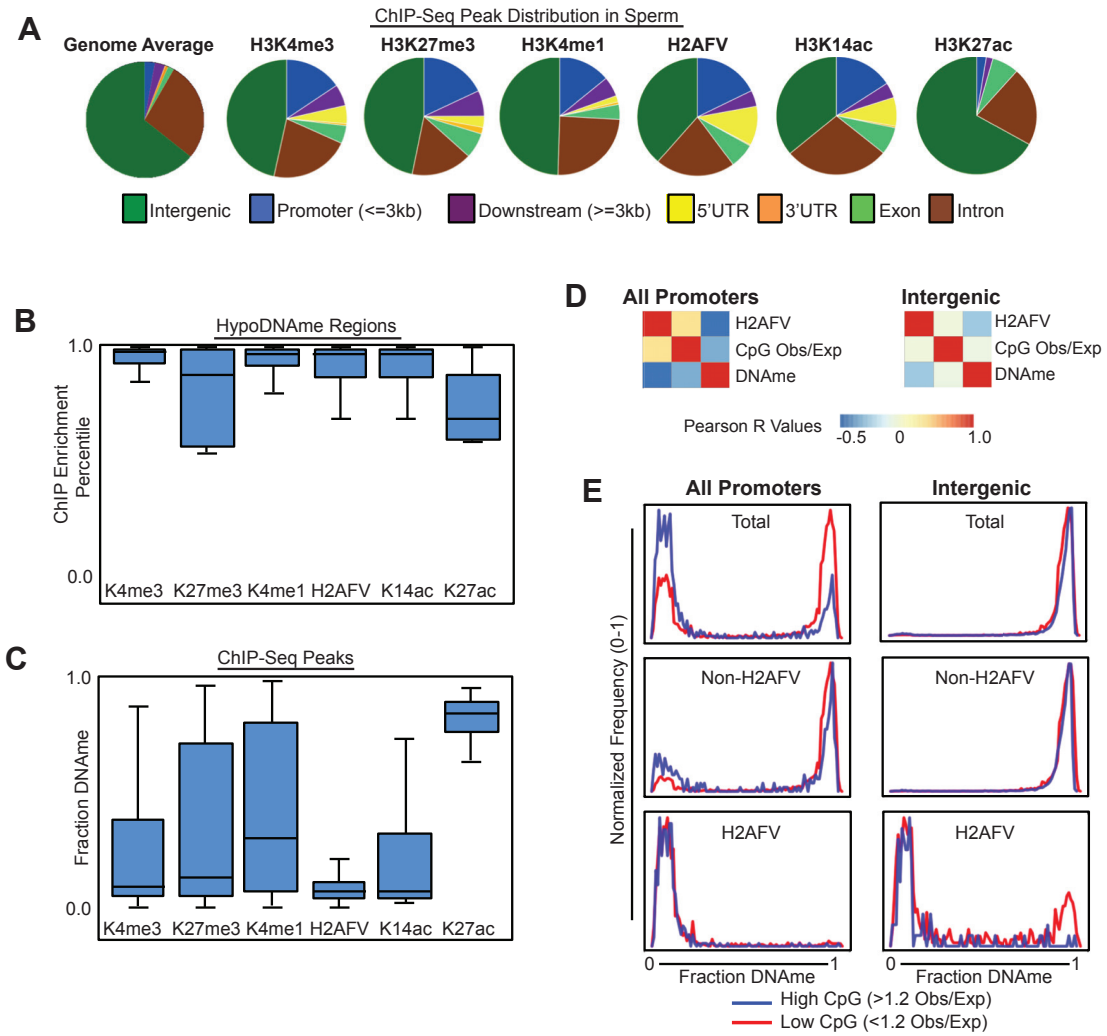


Figure S1. H2AFV and H3K4me1 Mark DNA Hypomethylated Regions in Zebrafish Sperm, Related to Figure 1

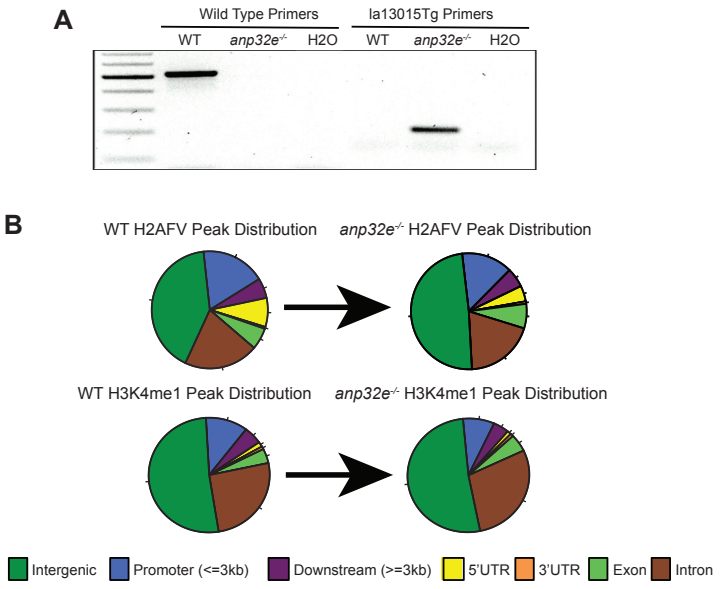
(A) Distribution of ChIP-Seq peaks in sperm assaying for various chromatin marks across a series of genomic landmarks. Non-enriched genome wide distribution is displayed for comparison.

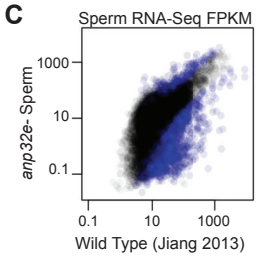
(B) ChIP enrichment percentiles for various chromatin marks at hypoDNAme regions in sperm.

(C) DNA methylation level (fraction) for enriched regions assays by ChIP-Seq in sperm.

(D) H2AFV is correlated with CpG density only at promoters. Pairwise correlation analysis as in (E), specifically focused on promoters (left) and intergenic loci (right).

(E) H2AFV marks hypoDNAme regions regardless of CpG density. Histograms indicate frequency after parsing regions for H2AFV, DNA methylation level, and CpG density. Left - promoters, and right - intergenic regions.





Genes	Ensembl Gene ID	Wild Type FPKM	Anp32e Null FPKM
dnmt3aa	ENSDARG00000005394	0.0	0.3
dnmt3ab	ENSDARG00000015566	0.0	0.1
dnmt1	ENSDARG00000030756	95.5	412.4
dnmt3bb.1	ENSDARG00000036791	13.1	36.7
dnmt3ba	ENSDARG00000052402	7.4	16.9
dnmt3bb.2	ENSDARG00000057830	8.4	27.0
dnmt3bb.3	ENSDARG00000057863	14.6	41.5
tet3	ENSDARG00000062646	0.0	0.7
tet1	ENSDARG00000075230	0.0	0.1
tet2	ENSDARG00000076928	0.0	0.0

Figure S2. Anp32e Null Sperm Have Mislocalized H2AFV, but Generally Not H3K4me1, Related to Figure 2

(A) Electrophoresis gel of PCR products from genotyping of la13015Tg insertion. Homozygous *anp32e* mutant fish have only the smaller mutant band and not the wild-type band. Primers indicated in Figure 2A were used in two separate PCR reactions per genotyped animal.

(B) Change in peak distribution (as in Figure S1A) for H2AFV and H3K4me1 in *anp32e* null sperm compared to wild-type.

(C) Comparison of *anp32e*- sperm RNA to a wild-type control (FPKM). Blue indicates greater than absolute 2-fold change in RNA ($p < 0.01$). Right - factors involved in DNA methylation regulation are listed as a table with FPKM values in mutant and wild-type sperm samples.

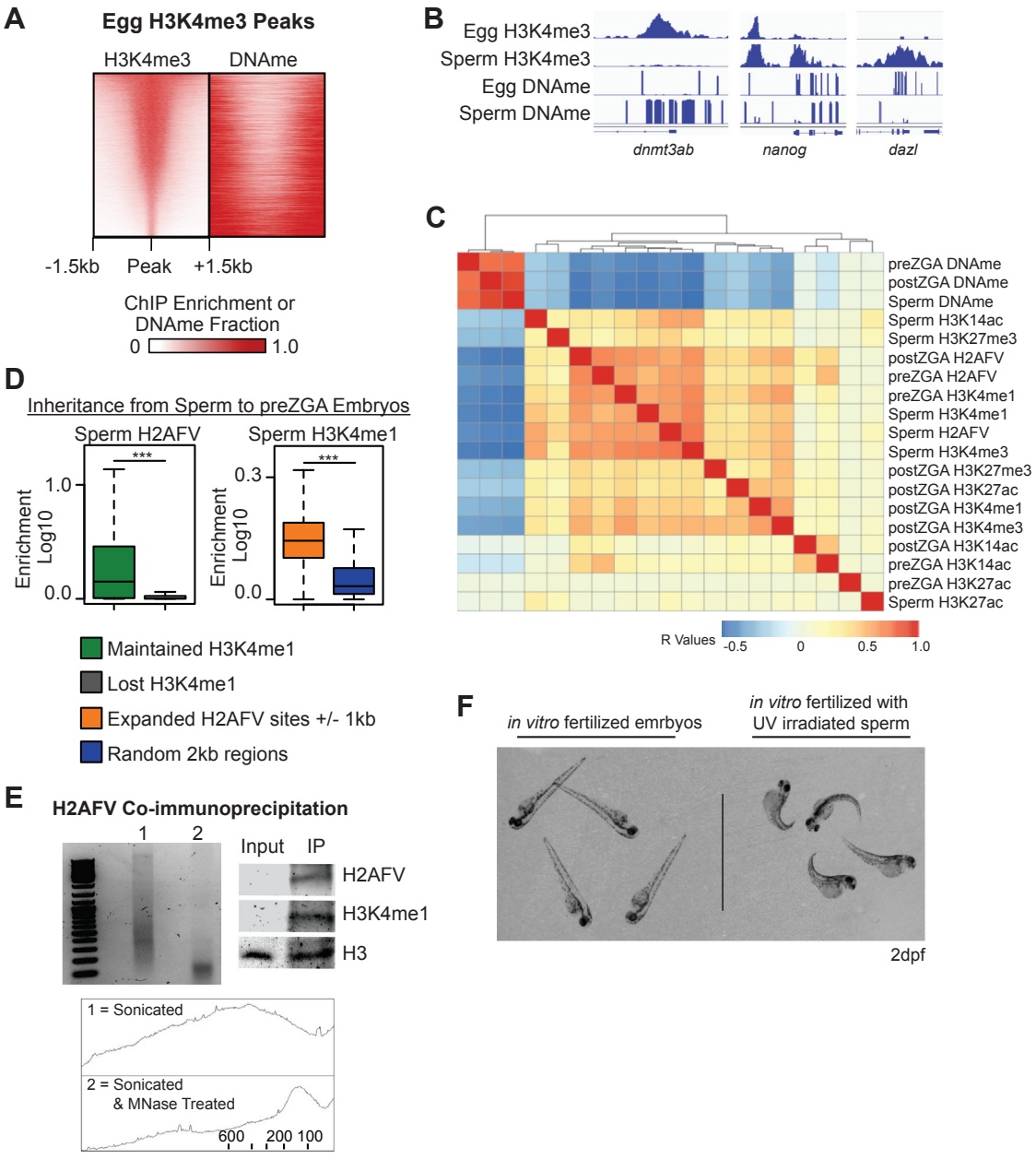


Figure S3. H2AFV and H3K4me1 Mark DNA Hypomethylated Regions in Zebrafish Cleavage Embryos, Related to Figure 3

(A) Heatmaps of H3K4me3 and DNA methylation at egg H3K4me3 peaks, indicating the two epigenetic marks rarely coincide. Zebrafish H3K4me3 egg data from Zhang et al., 2016.

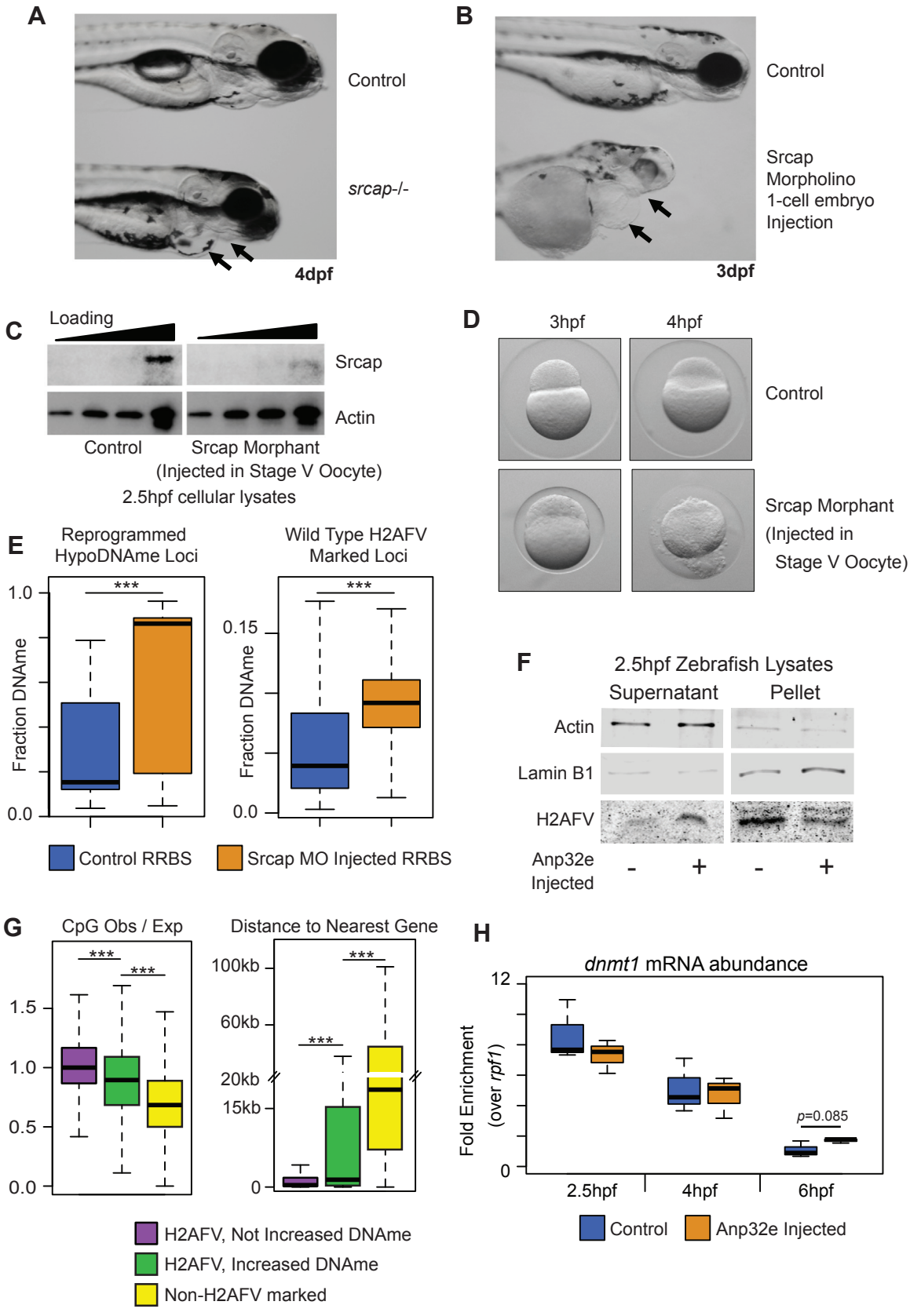
(B) Browser snapshots of enrichment for H3K4me3 and DNA methylation in sperm and eggs. H3K4me3 marks many promoters, but is absent from the promoters of transcription factor genes in eggs which also have moderate to high DNA methylation in eggs. Normalized read depth for ChIP-Seq data and DNA methylation fraction (0-1).

(C) Clustering of pairwise Pearson correlation R values for all epigenetic features assayed in sperm, and in pre- and post-ZGA embryo samples. H2AFV and H3K4me1 are most strongly correlated with one another and anti-correlated with DNAme.

(D) Maintenance of epigenetic marks during transition from sperm to preZGA embryos occurs at regions marked by H2AFV and flanked by H3K4me1 domains. Left – enrichment of H2AFV was measured at sites where H3K4me1 is lost compared to where it is maintained from sperm to embryos. Right – H3K4me1 is enriched where H2AFV expands in preZGA embryos as compared to control loci. H2AFV flanking regions were defined as peaks ± 1kb. t tests: *** = $p < 0.001$.

(E) H3K4me1 co-immunoprecipitates with H2AFV in preZGA embryos. Left - the DNA fragment size distribution for chromatin from preZGA embryos indicated that primarily mononucleosomes were used. Both an agarose gel image and the quantification of fragment size distribution (x axis = bp, and y axis = frequency) by ImageJ (bottom) are displayed. Top Right - western blot for H2AFV and H3K4me1 with H3 as a loading control.

(F) Gynogenetic embryos at 2 days post fertilization (dpf) compared to control embryos, which were generated in parallel using non-irradiated sperm. 100% of embryos at 2dpf displayed the expected phenotype (n > 30).



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Figure S4. Srcap Reduction Coincides with Increased DNA Methylation, Related to Figure 4

(A) Phenotype of zygotic null *srcap*$^{-/-}$ mutant fish compared to wild-type at 4dpf. Arrows indicate recessed jaw, and enlarged heart in mutant embryos.

(B) Similar to (A), except phenotypic characterization of embryos injected with an Srcap morpholino. Embryos are at 3dpf. As in (A), arrows indicate a recessed jaw and enlarged heart.

(C) A western blot for Srcap abundance comparing cellular lysates from control 2.5hpf embryos to lysates from embryos that resulted from *in vitro* fertilization of oocytes injected with an *srcap* morpholino.

(D) *srcap* morphants from stage V oocyte injection are modestly impaired at 3hpf, but have gross morphological defects at 4hpf - immediately following zygotic genome activation.

(E) RRBS-Seq DNA methylation analysis of *srcap* morphants compared to control embryos for loci which undergo DNA demethylation on the maternal allele under wild-type conditions (left), and for all wild-type H2AFV marked loci (right).

(F) Western blots indicate H2AFV decreases in the nuclear fraction and increases in the cytoplasmic fraction of cells purified from embryos that were injected with Anp32e – as in Figure 4F.

(G) Measurements of CpG density (left) and proximity to the gene promoters (right) for regions parsed by changes in DNA methylation after Anp32e injection. Regions were either marked by H2AFV in WT (violet and green) or lacked H2AFV (yellow). H2AFV marked loci were further parsed based on whether DNA methylation increased (green). t tests: *** = $p < 0.001$

(H) At 6hpf we observed a modest increase in *dnmt1* expression resulting from Anp32e injection. This was not the case at 2.5 or 4hpf. Overall *dnmt1* expression decreased during progression of cleavage phase.

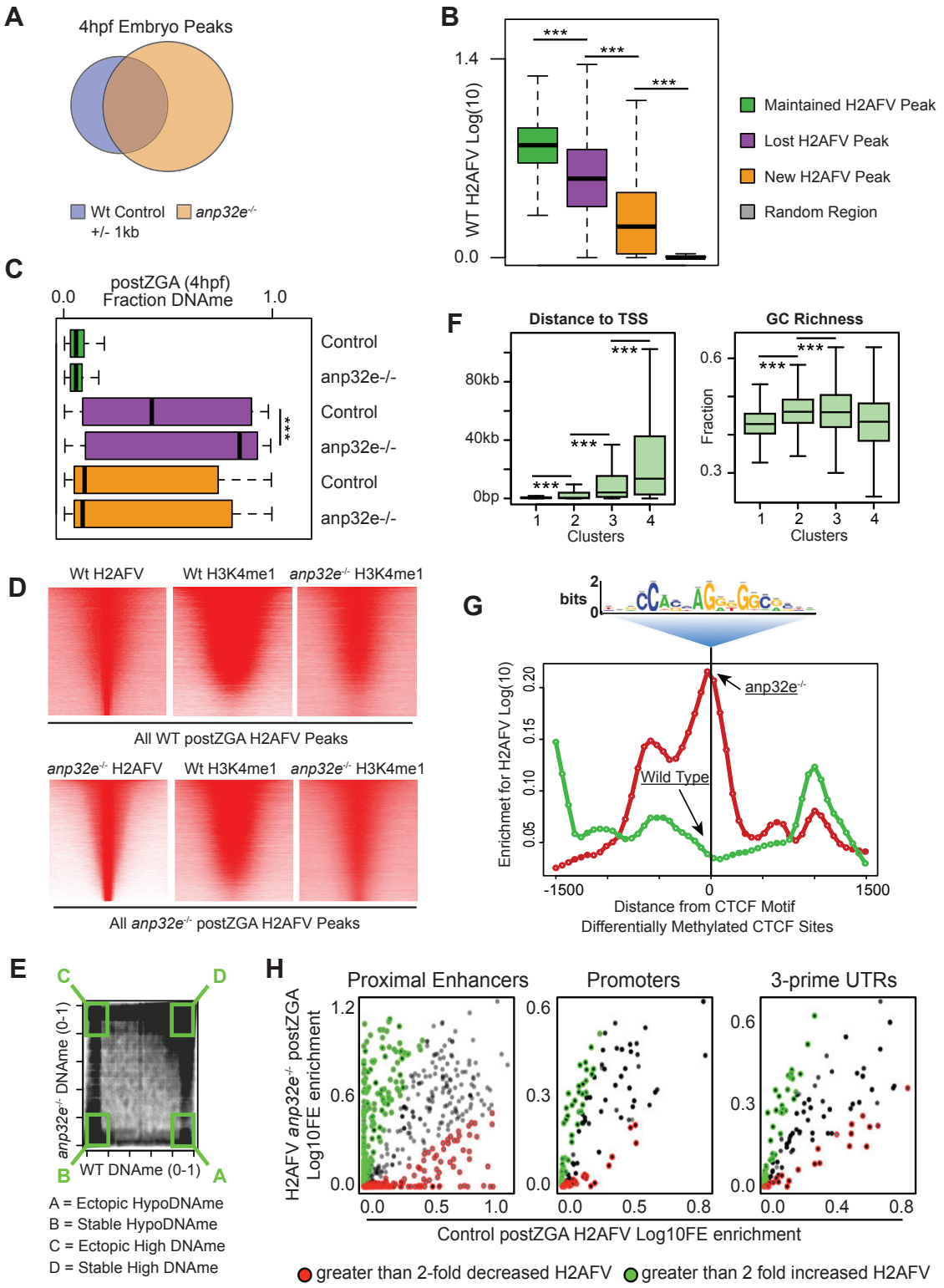


Figure S5. At Locations where H2AFV Increases, DNA Methylation Decreases, and Vice Versa, Related to Figure 5

(A) Venn diagram indicates the overlap of H2AFV peaks called in mutant samples compared to wild-type peaks ± 1kb.

(B) A comparison of H2AFV enrichment at loci where H2AFV changes in *anp32e*$^{-/-}$ embryos. Locations where H2AFV is robust in WT are unaffected by the *anp32e* mutation, while intermediate and low H2AFV sites are more susceptible.

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(C) DNAme levels significantly increase when H2AFV was lost in *anp32e*$^{-/-}$ embryos. Genomic loci were parsed as in (B). DNA methylation increased when H2AFV was lost, but because most increases in H2AFV (for *anp32e*$^{-/-}$ embryos) occurred proximal to site of moderate H2AFV in WT, these regions already lacked DNA methylation and were less affected.

(D) ChIP-Seq for H3K4me1 in wild-type and *anp32e*$^{-/-}$ embryos. Enrichment is similar for both wild-type and mutant H2AFV peaks, indicating that changing H2AFV has little effect on H3K4me1.

(E) A plot of mutant *anp32e*$^{-/-}$ DNA methylation at all scoreable CpGs as compared to wild-type. Green boxes and letters are used to highlight which CpGs correspond to regional H2AFV analysis in Figure 5C, and descriptions of highlighted boxes are indicated below.

(F) H2AFV increases at promoter-distal loci which have high GC content. As in Figure 5D, clusters from Figure 5A were analyzed for GC richness and distance to nearest transcription state site (TSS). t tests *** = $p < 0.001$.

(G) H2AFV is significantly higher in the *anp32e*$^{-/-}$ embryos than in the wild-type control embryos at differentially methylated CTCF sites ($p < 0.001$). CTCF motifs were identified using RSAT analysis applied to regions from Figure 5A cluster 3. These motifs were then mapped to the genome and CTCF motif containing regions were parsed based on DNA methylation change in the *anp32e* null embryos.

(H) Log10FE for H2AFV was measured for previously described ChIP-Seq data (Figure 5) in *anp32e*$^{-/-}$ mutants at three types of genomic features within *hox* gene regions (top GO-term). Feature types were proximal enhancers (marked by H3K27ac and H3K4me1 and within 5kb of a gene), promoters, or 3′-UTRs. These regions were then classified based on whether there was greater than a 2FC in H2AFV enrichment. Increasing H2AFV regions are marked in green and decreasing regions in red.

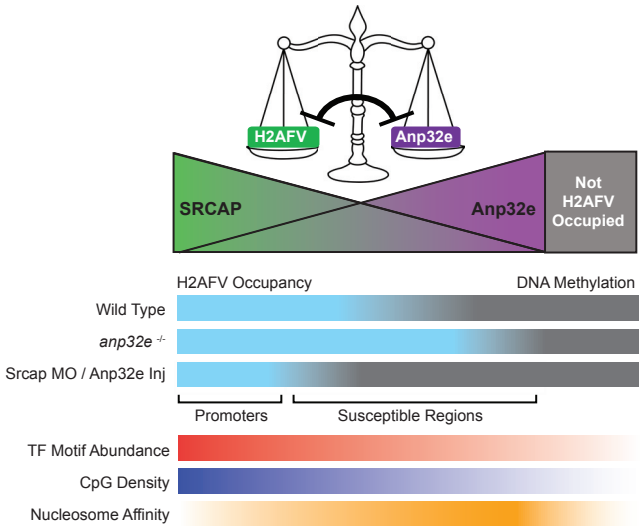


Figure S6. Balancing Placeholder Incorporation with Removal Facilitates DNA Methylation Patterning, Related to Figure 7

A model that considers the factors mediating dynamic Placeholder installation and removal, and the impact on DNA methylation when Placeholder is mis-localized. A key feature is the antagonism between the SRCAP complex, which deposits H2AFV - and the Anp32e chaperone, which removes H2AFV. Regulation of these antagonistic mechanisms determines Placeholder occupancy and therefore DNA methylation at dynamic H2AFV sites. First, we propose two modes of installation: 1) transcription factor-directed (with SRCAP complex), and 2) opportunistic (SRCAP complex-independent). Regions that lack TF sites, and have intrinsically low nucleosome affinity sequences are 'not H2AFV occupied' (gray region). The transcription factor-directed mode relies on the presence of TF binding motifs (which tend to be present in CpG-rich regions in vertebrates) and interaction of those TFs with SRCAP complex to ensure continual installation, even in the presence of Anp32e. The opportunistic mode relies on lower TF density and the intrinsic preference of Placeholder nucleosomes to occupy high nucleosome affinity sites. Anp32e may function to remove Placeholder both at these opportunistic regions and throughout the genome, but the primary impact occurs at 'susceptible' regions, which have modest CpG density, and high nucleosome density, but insufficient TF motif density to always counter Anp32e mediated removal. Notably, locations of TF-directed occupancy (mainly promoters) tend to have relatively low nucleosome affinity, but high enough TF density and efficient SRCAP complex targeting to effectively incorporate Placeholder in a highly efficient manner. When the balance of incorporation and removal is disrupted, by perturbing Anp32e or SRCAP complex, as in our genetic/molecular approaches, Placeholder occupancy is affected and DNA methylation is impacted.